

Probing Layer-wise Alignment Depth in Multimodal Transformers

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Abstract

Multimodal transformers achieve cross-modal understanding by implicitly learning to align representations from different modalities such as vision and language. Dual-stream architectures utilize separate encoders for each modality before attending between hidden representations in later layers via cross-attention. While this architecture is flexible, the optimal depth for cross-modal correspondence in these architectures remains unclear. This thesis investigates the evolution of representational alignment by systematically inserting cross-attention modules at varying depths. We evaluate these configurations on three diverse benchmarks: Hateful Memes, MM-IMDb, and UPMC-Food101. To understand the internal dynamics, we quantify representational alignment using metrics such as Centered Kernel Alignment (CKA) and Singular Vector CCA (SVCCA), alongside an analysis of effective dimensionality.

Our experiments consistently show that Late Fusion outperforms other strategies. Mechanistic analysis reveals that Early Fusion induces major effective dimensionality reduction in the vision stream, permanently constraining the model’s capacity to extract rich semantic features. In contrast, Late Fusion preserves modality-specific high-dimensionality before integration. Furthermore, we find that the correlation between representational alignment and performance is task-dependent, being strongest for tasks with redundant modalities.

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Declaration

By my own signature I declare that I produced this work as the sole author, working independently, and that I did not use any sources and aids other than those referenced in the text. All passages borrowed from external sources, verbatim or by content, are explicitly identified as such.

Signed:  Date: 11.02.2026

Chapter 1

Introduction

Human perception is fundamentally multimodal, integrating multiple senses to understand the world. This naturally inspires researchers to develop artificial intelligence systems with the same capabilities. In fields like autonomous driving, medical data analysis, and robotics, systems processing vision, language, and audio are essential for understanding complex environments. The transformer architecture (Vaswani et al., 2017) and the inherent attention mechanism revolutionized machine learning and led to state-of-the-art (SOTA) performance in many tasks. Initially, transformers were utilized for natural language processing and sequential data modeling. Building on this success, transformers were later adapted for computer vision tasks, achieving SOTA performance (Dosovitskiy et al., 2021) and demonstrating the architecture’s versatility with minimal inductive bias. In vision-and-language (V&L) research, recent developments in natural language processing and computer vision have converged, enabling deep learning models to operate on both modalities. Several approaches using the transformer architecture have emerged for aligning modalities. One approach uses separate encoders to transform the modalities in the first layers before they are integrated in later layers. Dual-stream architectures like ViLBERT (Lu et al., 2019) use several layers of cross-attention to achieve alignment between the vision and language modalities.

However, the optimal placement of cross-attention for alignment in dual-stream transformers remains unclear. The configuration of cross-attention layers is critical, as it determines how and when the modalities interact. The embeddings from each modality are inherently distinct in structure and information content and therefore need a sophisticated combination of self- and cross-modal transformation. If fusion happens too early, the modalities may not be sufficiently transformed to extract coherent features; if it occurs too late, the model may fail to capture complex cross-modal relationships. This thesis investigates alignment through the systematic insertion of cross-attention modules at varying depths.

Additionally, we quantify alignment using representational similarity metrics like Centered Kernel Alignment (CKA) (Kornblith et al., 2019). To our knowledge, representational alignment has not been systematically measured in dual-stream architectures. While representational alignment has only recently been used to analyze multiple modalities (e.g. Huh et al., 2024; Tjandrasuwita et al., 2025), prior work mainly focused on single-stream architectures. Representational similarity metrics like CKA can discover correspondences in the high-dimensional representations. Through layer-by-layer analysis of representational alignment across multiple datasets, this thesis systematically investigates how cross-attention placement influences both alignment evolution and downstream task performance.

Research Questions: This work analyzes representational alignment in dual-stream multimodal transformers, systematically investigating:

- RQ1: How does cross-attention placement influence the evolution of representational alignment across layers?
- RQ2: What is the relationship between representational alignment metrics and downstream task performance?
- RQ3: What representational changes occur when cross-attention is introduced at different depths?
- RQ4: How do alignment patterns vary across tasks with different modality characteristics?

The thesis is structured as follows: Chapter 2 introduces multimodal learning theory, transformer architectures, and the theoretical framework for representational alignment analysis using four complementary metrics (CKA, mkNN, SVCCA, OP). Chapter 3 describes the datasets, model configurations, training procedures, and analysis methods. Chapter 4 presents results from eight cross-attention configurations evaluated across three multimodal datasets, analyzing how placement depth influences representational alignment and downstream task performance. Chapter 5 summarizes the findings and discusses limitations.

Chapter 2

Background

2.1 Multimodal Learning

Humans perceive the world through multiple senses simultaneously. The signals we get from our senses are heterogeneous. For example, while the eyes sample visual signals from the world, our ears capture auditory signals. The brain collects these signals and combines them into an abstract model of the world. In this analogy, the senses represent distinct modalities.

Animals perceive the world through vastly different sensory systems. Dogs see in shades of blue and yellow, not red and green like humans. Bats navigate using echolocation, creating a sonic map of their environment representing a fundamentally different model. All of these different sensory systems encode information about the same world, but are adapted to their evolutionary needs.

To further build intuition, consider a concrete example. When you encounter a tree, you are simultaneously processing visual information (shape, color), sounds (rustling of leaves), and smells (pine scent). The brain integrates these pieces of information and forms a unified concept. Each sense captures information not available in the others. Yet there is also shared information across senses that the brain integrates to form a coherent perception. The brain leverages this mutual information to create integrated perceptions.

This notion of our own perception has sparked interest in the machine learning community to combine different modalities to achieve a holistic understanding of the world. A modality is simply a specific way of recording data from the world into a digitized format. In the tree example, one could capture an audio recording of the leaves rustling in the wind, capture a picture of the tree and describe it in natural language. These modalities here have different data formats, but convey shared information.

The central challenge in multimodal learning is to align these heterogeneous representations. In general, this is difficult, as modalities are fundamentally different. For instance, vision and language have distinct properties. Vision data are continuous and spatial, with a 2D array of pixels, while language is a sequence of discrete tokens. Simply adding them together does not work for such complex structures. Instead, the model learns to identify which aspects of the vision representations interact with which parts of the text, and vice versa. The task of combining these fundamentally different representations lies at the heart of multimodal learning. This enables richer representations that capture both shared information and each modality's unique information.

With the rise of encoder models such as word2vec (Mikolov et al., 2013) and later BERT (Devlin et al., 2019) vectorization of data has become popular. The core idea is to utilize a function

$$\text{enc} : \mathbb{R}^n \rightarrow \mathbb{R}^m \quad (2.1)$$

that maps the data points to a vector space of dimensionality m . During training, the matrix $\mathbf{W}$ and vector $\mathbf{b}$ are learned, such that the vector for a data point is calculated using $\mathbf{z} = \mathbf{W}\mathbf{x} + \mathbf{b}$. The high-dimensional vector $\mathbf{z}$ is referred to as the representation, latent representation or hidden representation. In this work, those terms are used interchangeably. The learned representation is then used for a downstream task like classification, regression or vector databases.

There are multiple benefits of having a learned representation. The representations enable downstream tasks. Instead of training models on raw data like pixels or words, more modular models are trained on these learned representations. Importantly, learned representations have fixed dimensionality, which standardizes inputs with varying input sizes. Texts with varying lengths can be encoded into a fixed dimensionality. Therefore, short messages and long articles can be encoded into the same dimensionality m .

A major advantage is the ability to use mathematical operations on the vectors. For instance, the dot product between two representations measures similarity between them. The dot product of two vectors encoding similar concepts is high, while for unrelated representations the value is small. This allows the model to organize knowledge based on semantics in large vector spaces. This part of machine learning is termed representation learning (Bengio et al., 2014).

However, real-world data often comes from multiple sources simultaneously. Rather than learning a representation from a single modality, multimodal learning extends this idea. Two or more different data formats are encoded into a shared representation that captures information from all modalities. This process of combining modalities into a unified representation is called fusion (Liang et al., 2023). Therefore, the goal of multimodal machine learning is to perform representation learning across multiple modalities, creating representations that are richer and more robust than any single modality alone.

2.1.1 Modality Theory

A modality in machine learning refers to a way data is recorded and stored in different formats. With the enormous corpora of data available on the internet, there is a myriad of data spread in various modalities. For example, videos on YouTube include relevant information that is not available in the corpus of natural language alone (Li and Tang, 2024). Current machine learning models utilize self-supervised learning to pretrain only on the language modality of the internet. Extending pretraining to other modalities prevalent in the web can increase performance significantly (Boehm et al., 2022).

In modality theory there is the distinction of raw and abstract modalities (Liang et al., 2023). This distinction is not strictly binary, but rather continuous. Raw modalities are signals directly recorded by sensors, like pixels from cameras, or pressure waves captured by microphones. They are high-dimensional and have idiosyncratic structure. By encoding the raw modalities using suitable encoders, the representations become more similar in structure, which is referred to as abstract modalities. This thesis focuses on abstract modalities, where each modality is encoded by a pretrained model (BERT for text and ViT (Dosovitskiy et al., 2021) for vision).

The literature distinguishes between homogeneous and heterogeneous modalities (Liang et al., 2023). Two samples from homogeneous modalities share structure and other qualities. Samples from heterogeneous modalities have inherently distinct characteristics. Homogeneous samples lead to similar representations, heterogeneous samples not. For example, vision and language are heterogeneous, as the data structure is different. Vision data are continuous and spatio-temporal, while natural language is discrete and sequential.

The distinction between heterogeneous and homogeneous modalities, in combination with mutual information, is illustrated in Figure 2.1.

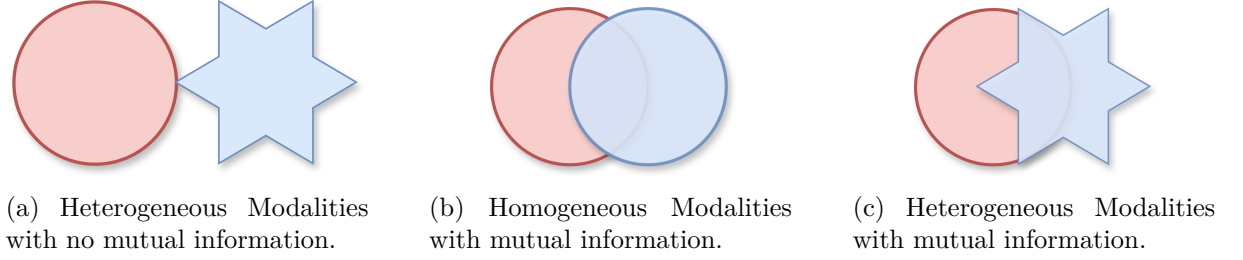


Figure 2.1: Heterogeneous vs. Homogeneous Modalities.

The term alignment refers to aligning mutual information and integrating distinct information from the heterogeneous data points. These theoretical foundations lead to two key properties that guide multimodal learning approaches (Li and Tang, 2024; Liang et al., 2023):

- *Different modalities are heterogeneous:* Information in modalities has different structures, amounts of noise, and varied distributions.
- *Modalities share some information* about an underlying ground truth. The mutual information can be aligned between them and further complement each other with new information.

Mutual information $I(X;Y)$ quantifies the shared information between two modalities: high mutual information indicates the modalities encode overlapping concepts, while low mutual information suggests they contain complementary information.

Mutual information between two random variables X, Y is defined as:

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = \mathbb{E}_{X,Y} \left[\log \frac{p(x,y)}{p(x)p(y)} \right] \quad (2.2)$$

where $H(X)$ and $H(Y)$ are the individual entropies, and $H(X,Y)$ is the joint entropy of the two variables. Using learning algorithms, this mutual information is detected and aligned across the distinct modalities.

2.2 Multi-Stream Architecture

Fusion of modalities is a non-trivial task. While there are explicit and implicit alignment methods (Liang et al., 2023), SOTA performance has been achieved with implicit methods using transformers. Transformers learn to align multiple modalities, with minimal inductive bias.

For transformers, different fusion methods have been proposed. The prominent approaches are single-stream and multi-stream alignment. Single-stream architectures like ViLT (Kim et al., 2021) concatenate embeddings before passing them through a transformer. Raw modalities are fused together and the single stream transformer extracts cross-modal interactions.

Multi-stream architectures, on the contrary, utilize separate encoders (Radford et al., 2021; Lu et al., 2019). They offer the advantage of flexibility. Each modality stream allows self-attention to develop more abstract features in one modality, while cross-attention acts as a coordination between both streams. Furthermore, each stream leverages pretrained unimodal encoders, which already extract relevant features. Therefore, representations from more abstract modalities are fused. This work focuses on dual-stream architectures, specifically a variant of ViLBERT.

2.2.1 Fusion Depth: Early, Middle, and Late Fusion

While dual-stream architectures provide flexibility, a critical design choice is the layer depth at which the two streams interact. Following Liang et al. (2023), this can be classified as early fusion (streams interact at shallow layers), middle fusion (interaction at intermediate layers), or late fusion (interaction at deep layers).

This taxonomy is visualized in Figure 2.2. Early fusion integrates raw or minimally processed modalities, while late fusion allows each stream to develop modality-specific features before interaction. This distinction has important implications: early fusion may constrain representational capacity by forcing interaction before abstract features emerge, while late fusion allows richer unimodal processing but risks missing early cross-modal constraints.

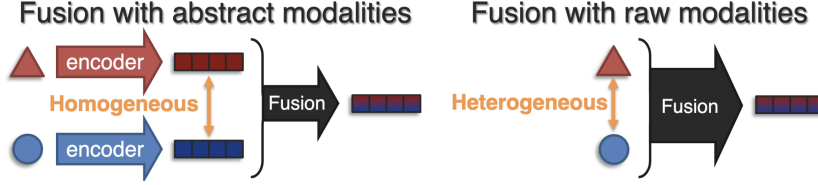


Figure 2.2: Distinction of modality fusion at different depths in the model. Left: Late fusion with encoded, more abstract and homogeneous modalities. Right: Early Fusion with raw, heterogeneous modalities (Liang et al., 2023).

In this thesis, we systematically vary fusion depth to understand how the layer at which cross-attention is inserted affects representational alignment and downstream performance.

2.3 Cross-Attention

Lu et al. (2019) proposed cross-attention modules at intermediate depths to achieve cross-modal alignment during pretraining. This extends the single-stream BERT architecture, where only self-attention is used.

Attention Mechanism

Cross-attention builds upon the Scaled Dot-Product Attention mechanism introduced by Vaswani et al. (2017). The transformer computes three projections of the input representing Queries, Keys, and Values, using trainable matrices $\mathbf{W}_Q$, $\mathbf{W}_K$, $\mathbf{W}_V$:

$$\begin{aligned} Q &= \mathbf{W}_Q \mathbf{X} \\ V &= \mathbf{W}_V \mathbf{X} \\ K &= \mathbf{W}_K \mathbf{X} \end{aligned}$$

The attention operation computes a weighted sum of the values, where the weights are determined by the compatibility between query and corresponding keys. This is formally defined as:

$$\text{attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right) \mathbf{V}.$$

The dot product $\mathbf{Q}\mathbf{K}^\top$ computes similarity scores between the query and keys. These scores are scaled by the factor $\frac{1}{\sqrt{d_k}}$ to prevent the dot products from growing too large in magnitude,

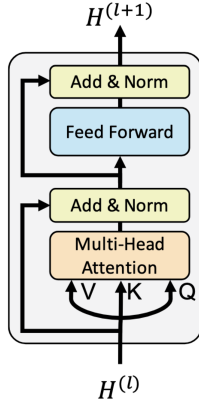
which would otherwise push the softmax function into regions with extremely small gradients, destabilizing training.

The softmax function is then applied to normalize these scaled scores, resulting in an attention distribution where values sum to 1. This distribution acts as a weighting mechanism. High similarity scores result in larger weights, allowing the model to attend more heavily to relevant parts of the input. Finally, these weights are used to compute a weighted aggregation of the Value matrix $\mathbf{V}$. Unlike a simple lookup that retrieves a single match, this mechanism allows the model to integrate information from the entire input sequence based on relevance.

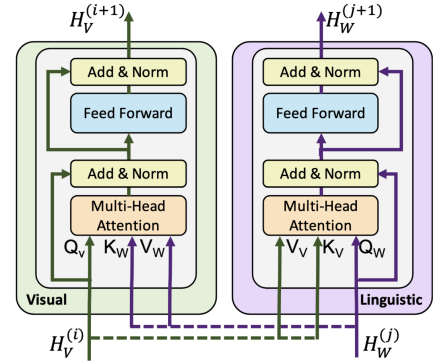
Self-Attention versus Cross-Attention

The key difference is the source of the query, key, and value projections. In self-attention, all three come from the same modality. In cross-attention, the query comes from one modality while keys and values come from another.

Self-attention transforms the input itself, while cross-attention integrates information from another modality. This mechanism is illustrated in Figure 2.3b.



(a) Original encoder block (Vaswani et al., 2017).



(b) ViLBERT cross-attention block (Lu et al., 2019).

In ViLBERT, cross-attention layers are strategically inserted between self-attention layers. The vision stream processes image tokens through self-attention, then cross-attends to the language stream. Simultaneously, the language stream self-attends, then cross-attends to vision. This bidirectional interaction allows the model to discover cross-modal correspondences during training. However, it is still unclear how alignment evolves across layers. This motivates the experiments in this thesis, where cross-attention modules are systematically inserted at different depths and their effect on representational alignment is evaluated.

2.4 Theoretical Framework

Alignment refers to identifying mutual information in different modalities and integrating them into a cross-modal representation. The dual-stream architecture used in the experiments utilizes cross-attention transformer blocks to achieve implicit alignment. In order to analyze at what depth cross-modal correspondence occurs, it is essential to have quantitative measures of alignment.

Multi-layer neural networks compute hidden representations as input for the subsequent layer. Representational analysis examines similarities in those latent representations. The analysis can be multifaceted with possible comparisons across different layers and architectures.

Kornblith et al. (2019) established a framework of a similarity function with Centered Kernel Alignment (CKA) to compare latent representations between different layers and models.

A similarity function is a function comparing two vectors of same dimensionality d :

$$\varphi : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$$

vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$ are hidden representations from two distinct layers, models or modalities. The function $\varphi(\mathbf{u}, \mathbf{v})$ performs a similarity comparison and results in a scalar value. Ideally, the resulting scalar is bounded by $[0, 1]$, but as will be discussed below, this is not the case for all metrics.

For comparing high-dimensional representations, specific invariances are mandatory. Invariance in a function is defined as follows. Let $f : X \rightarrow Y$ be a function and $\odot$ be an operation on the set X . f is invariant to operation $\odot$, if the following holds:

$$f(x) = f(x \odot a), \forall x, a \in X$$

For example: For representations $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, where $\mathbf{v} = -\mathbf{u}$, the representations are structurally identical, but reflected. It is necessary for similarity functions to be invariant to such transformations.

The following four properties are critical for similarity functions, as shown by Klabunde et al. (2025); Kornblith et al. (2019):

1. Invariance to orthogonal transformations: For orthogonal matrices $\mathbf{A}, \mathbf{B}$ (where $\mathbf{A}^\top \mathbf{A} = \mathbf{I} \Rightarrow \mathbf{A}^{-1} = \mathbf{A}^\top$), the similarity score remains unchanged: $\varphi(\mathbf{A}\mathbf{u}, \mathbf{B}\mathbf{v}) = \varphi(\mathbf{u}, \mathbf{v})$
2. Invariance to isotropic scaling: For scalars $c, d \in \mathbb{R}^+$, uniform scaling preserves similarity: $\varphi(c\mathbf{u}, d\mathbf{v}) = \varphi(\mathbf{u}, \mathbf{v})$
3. Invariance to translations: Two representations $\mathbf{u}, \mathbf{v}$ are similar if they are only translated by a vector. This enables comparisons of structures of representations that are shifted in space (have a different mean) (Klabunde et al., 2025, 2023).

Note here that other invariances are also applicable for specific problems, as discussed extensively in Klabunde et al. (2025). As different metrics have different invariances, the choice of similarity function is critical. The objective is to have a range of similarity measures with different invariances to have a robust analysis.

Global vs. Local Alignment: Representational alignment can be analyzed at two distinct scales (Liang et al., 2023):

Global Alignment measures the structural similarity of the entire representation space. Given two modality matrices $\mathbf{X}, \mathbf{Y}$, metrics like CKA, SVCCA, and Procrustes assess whether the global geometry (the overall "shape" of the data cloud) is preserved across modalities.

Local Alignment focuses on the preservation of neighborhood topology. In this work, we use mutual k-nearest neighbors (mkNN) to determine if samples that are close in the vision space are also close in the text space.

To visualize the difference, consider the concept of "dog." Both modalities might cluster dog instances tightly together (high local alignment). However, the position of the "dog" cluster relative to "cat" or "frog" clusters might differ significantly between modalities (low global alignment). Local alignment captures the former, while global metrics assess the latter.

Beyond the scale of alignment, recent literature distinguishes between representational similarity (structural closeness of internal states) and functional similarity (similarity in model outputs and performance) (Klabunde et al., 2025). While prior work has often relied on probing

to assess functional capability (Frank et al., 2021), this thesis focuses primarily on representational similarity to analyze the internal mechanics of fusion. We use functional performance on downstream tasks primarily to validate these internal findings.

2.5 Alignment Measures

In the literature, several measures to compare neural networks were proposed. Klabunde et al. (2025) discuss a variety of representational similarity measures with respect to their invariances. Their work included a ranking of methods and proposed practical recommendations for usage in further studies. They found that representational similarities vary significantly between different measures (Klabunde et al., 2023, 2025). In the structural analysis of ViLBERT’s learned representations between modalities, we implemented several measurements, but this section mainly focuses on Centered Kernel Alignment (CKA) Kornblith et al. (2019), Singular Vector CCA (Raghu et al., 2017) mutual K Nearest-Neighbors (mkNN) and Orthogonal Procrustes (OP). Details on the metrics as well as a rationale for the metric selection are provided in Section 2.5.

Experiments on decoder-only transformers showed high variance in similarity scores for different measures, even with the same invariances (Klabunde et al., 2023). Some representations with high similarity scores on one measurement, resulted in low scores in another. Furthermore, scores are often hard to interpret, since a range of measures is not bounded. Evaluation on different tasks in Klabunde et al. (2023) indicated inconsistencies in similarity implying task dependence in alignment. The research in the above paper was conducted on decoder-only models, modern Large Language Models. Our research extends this to a different paradigm of encoder-only models, such as BERT. In addition, autoregressive LLMs have the property of being indeterministic, while this is not the case for encoder-only models. The field of systematically analyzing the models is still in its infancy and requires further research. Research suggests a combination of different measurements to have a robust analysis (Klabunde et al., 2025, 2023).

Centered Kernel Alignment

Centered Kernel Alignment (Kornblith et al., 2019) provides a robust global similarity measure between two representations.

Let $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n \times d}$ be mean-centered activations of two models to compare, where n is the number of samples and d the dimensionality of the representation. Row $\mathbf{X}_{i,:}$ represents the activations of a single sample. Therefore, for a fixed i , $\mathbf{X}_{i,j}$ are the features in an example.

$$\langle \text{flatten}(\mathbf{X}\mathbf{X}^\top), \text{flatten}(\mathbf{Y}\mathbf{Y}^\top) \rangle = \text{tr}(\mathbf{X}\mathbf{X}^\top \mathbf{Y}\mathbf{Y}^\top) = \|\mathbf{Y}^\top \mathbf{X}\|_F^2 \quad (2.3)$$

where $\langle \cdot, \cdot \rangle$ is the inner product and $\|\cdot\|_F$ is the Frobenius norm.

Here $\mathbf{X}\mathbf{X}^\top$ computes the matrix of sample similarities. Specifically, this computes pairwise sample similarities: for each sample $\mathbf{X}_{i,:}$, we compute its similarity to all other samples in $\mathbf{X}$:

$$\mathbf{X}_{i,:} \cdot \mathbf{X}_{j,:}, \forall i, j \quad (2.4)$$

For centered feature matrices, the Hilbert-Schmidt Independence Criterion (HSIC) (Gretton et al., 2005) relates to the covariance between features:

$$\text{HSIC}(\mathbf{X}\mathbf{X}^\top, \mathbf{Y}\mathbf{Y}^\top) = \frac{1}{(n-1)^2} \text{tr}(\mathbf{X}\mathbf{X}^\top \mathbf{Y}\mathbf{Y}^\top) = \|\text{cov}(\mathbf{X}^\top, \mathbf{Y}^\top)\|_F^2 \quad (2.5)$$

Centered Kernel Alignment (CKA) with linear kernel normalizes HSIC to be invariant to isotropic scaling:

$$CKA(\mathbf{X}, \mathbf{Y}) = \frac{HSIC(\mathbf{X}\mathbf{X}^\top, \mathbf{Y}\mathbf{Y}^\top)}{\sqrt{HSIC(\mathbf{X}\mathbf{X}^\top, \mathbf{X}\mathbf{X}^\top) HSIC(\mathbf{Y}\mathbf{Y}^\top, \mathbf{Y}\mathbf{Y}^\top)}} \quad (2.6)$$

CKA measures the structural similarity between two representations while being invariant to orthogonal transformations and isotropic scaling. The metric is bounded by $CKA(\mathbf{X}, \mathbf{Y}) \in [0, 1]$ where 1 means perfect similarity and 0 means no similarity. We choose this metric, as it is a popular metric in recent literature (Klabunde et al., 2025; Kornblith et al., 2019), where CKA with different kernels were studied. This thesis applies CKA with linear kernel to compare text and image representations at each layer depth.

Mutual k-Nearest Neighbors (mkNN)

Mutual k-Nearest Neighbors (Huh et al., 2024) provides a local alignment measure that complements the global perspective of CKA. The method builds on the general principle of nearest-neighbor approaches: for a given representation of a single sample $\mathbf{u} \in \mathbb{R}^d$, we compare it to all other representations using a similarity function and identify the k most similar representations. mkNN extends this concept to the multimodal setting. Given two representation sets $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{n \times d}$ with n samples of dimensionality d from different modalities, mkNN measures how often samples that are k -nearest neighbors in one modality are also k -nearest neighbors in the other modality. This captures local alignment by focusing on whether corresponding samples maintain similar neighborhood structures across modalities.

For a pair of corresponding samples $\mathbf{u} = \mathbf{X}_{i,:}$ and $\mathbf{v} = \mathbf{Y}_{i,:}$ from the two modalities, we first identify their sets of nearest neighbors in their respective representations:

$$\begin{aligned} Z_{\mathbf{X}, \mathbf{u}} &:= \text{kNN}(\mathbf{u}, \mathbf{X} \setminus \{\mathbf{u}\}) \\ Z_{\mathbf{Y}, \mathbf{v}} &:= \text{kNN}(\mathbf{v}, \mathbf{Y} \setminus \{\mathbf{v}\}) \end{aligned} \quad (2.7)$$

where $Z_{\mathbf{X}, \mathbf{u}}$ denotes the set of k nearest neighbors of $\mathbf{u}$ in $\mathbf{X}$. The nearest neighbors are determined using cosine similarity via dot product on normalized features.

The mkNN metric then computes the average overlap of mutual nearest neighbors across all n sample pairs:

$$\text{mkNN}(\mathbf{X}, \mathbf{Y}) = \frac{1}{n} \sum_{i=1}^n \frac{1}{k} |Z_{\mathbf{X}, \mathbf{X}_{i,:}} \cap Z_{\mathbf{Y}, \mathbf{Y}_{i,:}}| \quad (2.8)$$

where $|\cdot|$ denotes set cardinality. This approach effectively constrains the alignment measurement to local neighborhoods, making it particularly suitable for analyzing cross-modal correspondence where only subsets of representations should align (Huh et al., 2024).

Orthogonal Procrustes

Orthogonal Procrustes (OP) (Ding et al., 2021) is a distance metric invariant to orthogonal transformations that captures geometric alignment.

$$\text{procrustes}(\mathbf{X}, \mathbf{Y}) = \min_{\mathbf{Q}} \|\mathbf{X}\mathbf{Q} - \mathbf{Y}\|_F \quad (2.9)$$

where $\mathbf{Q}$ is an orthogonal matrix satisfying $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$.

The intuition behind OP is to find an orthogonal transformation $\mathbf{Q}$ that minimizes the Frobenius norm of the differences between $\mathbf{X}\mathbf{Q}$ and $\mathbf{Y}$. As this metric is only sensitive to orthogonal transformations, isotropic scaling invariance can be achieved by normalizing both representations before calculating the OP score (Klabunde et al., 2023). OP has a lower bound

of 0 (perfect alignment) and an unbounded upper bound, as it measures the residual distance between representations. Being a dissimilarity measure, lower values indicate better alignment.

Correlation analysis showed no significant correlation to other similarity measures (Figure A.2), indicating that it captures different aspects of alignment.

Singular Vector Canonical Correlation Analysis

Singular Vector Canonical Correlation Analysis (SVCCA) (Raghu et al., 2017) is a combination of Singular Value Decomposition (SVD) and Canonical Correlation Analysis (CCA) (Hardoon et al., 2004) used for comparing neural networks.

The computation of the metric consists of two stages: In the first step, SVD is utilized to reduce dimensionality of the hidden representations. This removes noise from high-dimensional representations that are susceptible to small perturbations. For representations $\mathbf{X} \in \mathbb{R}^{n \times d}$ and $\mathbf{Y} \in \mathbb{R}^{n \times d}$, SVD decomposes each as $\mathbf{X} = \mathbf{U}_\mathbf{X} \mathbf{\Sigma}_\mathbf{X} \mathbf{V}_\mathbf{X}^\top$ and $\mathbf{Y} = \mathbf{U}_\mathbf{Y} \mathbf{\Sigma}_\mathbf{Y} \mathbf{V}_\mathbf{Y}^\top$. Keeping only the top k singular vectors (equivalent to PCA) yields denoised representations $\tilde{\mathbf{X}} = \mathbf{U}_\mathbf{X}[:, :k]$ and $\tilde{\mathbf{Y}} = \mathbf{U}_\mathbf{Y}[:, :k]$.

Second, CCA is used to find correlations between the reduced representations $\tilde{\mathbf{X}}$ and $\tilde{\mathbf{Y}}$.

CCA is computed by maximizing correlation between linear projections of the two representations.

$$\rho_i(\tilde{\mathbf{X}}, \tilde{\mathbf{Y}}) = \max_{a_i, b_i} \text{corr}(a_i^\top \tilde{\mathbf{X}}, b_i^\top \tilde{\mathbf{Y}}) \quad (2.10)$$

where ρ_i is the i -th canonical correlation. The final SVCCA score is the mean of these canonical correlations.

$$\text{SVCCA}(\mathbf{X}, \mathbf{Y}) = \frac{1}{k} \sum_{i=1}^k \rho_i(\tilde{\mathbf{X}}, \tilde{\mathbf{Y}}) \quad (2.11)$$

SVCCA is invariant to translations, permutations, orthogonal transformations and isotropic scaling (Klabunde et al., 2025). The metric is bounded by $[0, 1]$.

Metric Idiosyncrasies

CKA, mkNN, SVCCA and OP have different properties and invariances, capturing distinct aspects of alignment. While CKA is a global alignment metric for structural similarity, mkNN employs local neighborhoods to assess alignment. SVCCA covers different aspects of alignment through dimensionality reduction. OP belongs to the class of alignment-based measures (Klabunde et al., 2025), where a transformation $\varphi \in \mathbb{T}$ is found that minimizes $\|\varphi(\mathbf{X}) - \mathbf{Y}\|$. Unlike similarity measures that compute statistical relationships, OP directly finds the optimal orthogonal transformation between representation spaces. This geometric perspective captures different aspects of alignment than the statistical approaches of CKA, mkNN, and SVCCA, as evidenced by its low correlation with other metrics (Figure A.2).

The combination of the four metrics provides a more robust analysis of representational alignment than any single metric alone. Table 2.1 summarizes the invariance properties of each metric, demonstrating their complementary nature for robust representational alignment analysis.

2.6 Related Works

Recent work systematically compared single- and dual-stream architectures under controlled settings (Bugliarello et al., 2021). Their experiments indicated no significant performance difference between architectural families on various V&L tasks when trained with identical data and hyperparameters.

Metric	OT	IS	T
CKA	✓	✓	✓
mkNN	✓	✓	–
SVCCA	✓	✓	✓
OP	✓	–	✓

Table 2.1: Invariance properties of alignment metrics. OT: Orthogonal Transformations, IS: Isotropic Scaling, T: Translation.

Guo et al. (2023) further investigated fusion depth for dual-stream architectures like ViLBERT. Introducing a framework to dynamically adjust fusion depth per sample, they found that this dynamic approach outperforms static fusion. However, their analysis focused on task-specific architectural patterns rather than quantitative representational alignment.

While these studies analyzed performance implications of fusion depth, they did not systematically analyze representational alignment across layers using similarity metrics. This thesis investigates alignment quantitatively through layer-wise analysis.

Analyzing alignment in multimodal transformers requires quantitative measures of representational similarity. While some work implicitly analyzes multimodality through probing at inference time (Frank et al., 2021), systematic analysis of representational alignment in dual-stream architectures remains limited.

Metrics for comparing hidden representations were formalized by Kornblith et al. (2019) with Centered Kernel Alignment (CKA). Their work established theoretical foundations for similarity functions, including required invariance properties for meaningful representation comparisons. Since then, CKA has been utilized for various analyses of neural networks. For example, Ramasesh et al. (2020) incorporated CKA to analyze catastrophic forgetting.

Building on this, Huh et al. (2024) proposed the Platonic representation hypothesis: with sufficiently capable encoders, independent unimodal models should learn similar representations across modalities. They tested this by computing CKA and other similarity measures across different modalities and architectures using single-stream encoders. The hypothesis was motivated by the multitask scaling hypothesis—larger models with more parameters optimize over smaller solution spaces, pushing independent learners toward similar representations.

However, Tjandrasuwita et al. (2025) found empirical limitations to this hypothesis. Using CKA on CLIP models, they showed that vision-language representations remain quite dissimilar. Critically, they found that the Platonic hypothesis holds only for redundant modalities with high mutual information; tasks with modality-specific information show weak and unstable alignment correlations.

Despite these advances, no prior work has systematically analyzed representational alignment in dual-stream architectures like ViLBERT, where cross-attention placement and its effect on alignment across layers remains unexplored. This gap is the focus of the present work.

Chapter 3

Methodology

3.1 Datasets

For the pretraining process, the architecture was trained on a subset of the Conceptual Captions dataset (Sharma et al., 2018). The collection contains around 3.3 million image-caption pairs scraped from the internet. These captions were generated using an alt-text algorithm, so the quality of the captions is noisy. However, the data corpus is large and diverse, making it suitable for pretraining multimodal models. The large scale and diversity are particularly important for multimodal architectures, as they enable the model to learn robust cross-modal correspondences across varied visual-textual contexts. While the encoders are pretrained on even larger sets, the diverse multimodal set allows the models to learn effective cross-modal alignment between vision and language representations. After data cleansing, the dataset shrank to around 2 million samples. For different training runs, a randomized subset of 500k image-text pairs was used, as the hardware was limited. For fine-tuning, three established multimodal benchmark datasets were used: Hateful Memes (Kiela et al., 2021), UPMC-Food101 (Wang et al., 2015), and MM-IMDb (Arevalo et al., 2017). These datasets have been widely adopted in multimodal alignment research (Ma et al., 2022; Tjandrasuwita et al., 2025; Radford et al., 2021).

Hateful Memes (Kiela et al., 2021) is a dataset containing humorous images with their captions. The objective of the dataset is to distinguish between hateful memes and non-hateful ones. Hateful memes contain offensive and/or discriminating jokes. The challenge in this dataset is the complex nature of humor, which often requires understanding both the visual and textual modalities to grasp the intended meaning. For example, the text might be clearly offensive and harmful, but only in combination with the image is the sarcastic and humorous nature revealed. Figure 3.1 shows two examples from the dataset, where the same caption is used with different images, leading to different classifications. This data collection is well suited for multimodal models, as hateful memes require both modalities to be understood. In addition, cross-modality correspondence is more complex than simple caption matching, as humor has complex relationships between the vision and language modalities. The train set consists of 8500 samples with a slight class imbalance of 55% positive labels, the rest negative. To account for this imbalance, a weighted Binary Cross Entropy (BCE) loss function was used during training. To validate results, we checked performance on the validation set with 1040 samples after every epoch and reported final results on the test set, which consists of 3000 samples. The weighted loss function was only used during training; in validation and test phases, an unweighted BCE was used.

UPMC-Food101 (Wang et al., 2015) is a dataset with images of food dishes and their recipes. The task is to classify the dish into one of 101 categories. The dataset contains around 90k



(a) Non-hateful example



(b) Hateful example (stereotype pairing)

Figure 3.1: Examples from Hateful Memes dataset illustrating the necessity of multimodal understanding. The same caption combined with different images produces fundamentally different classifications (Kiela et al., 2021)

samples, split into train (64k), validation (16k), and test (9k) sets. Wang et al. (2015) showed that multimodal models outperform unimodal baselines significantly, indicating that the dataset is well suited for multimodal learning.

MM-IMDb (Arevalo et al., 2017) is a dataset that contains movie plots and posters, with the task of predicting the genre of the movie. The training set includes around 20k samples, with a multi-label classification task of 23 genres. The sizes of the validation and test sets are about 5k each.

We categorize these datasets by difficulty based on modality redundancy. UPMC-Food101 is considered the easiest task due to high redundancy between recipes and food images. MM-IMDb is moderately difficult, requiring correlation between posters and plot genres. Hateful Memes is the most difficult, as it requires resolving non-redundant, often contradictory (sarcastic) information between text and image.

Dataset	Task Type	Train	Val	Test
Conceptual Captions	Pretraining	500k	11k	/
Hateful Memes	Binary Classif.	8.5k	1k	3k
UPMC-Food101	Multi-class Classif.	64k	16k	9k
MM-IMDb	Multi-label Classif.	20k	5k	5k

Table 3.1: Dataset statistics and prediction tasks.

During the process of this thesis, the models were trained on the train splits and evaluated on the validation split for epoch-evaluation and early stopping. The final results were reported on the test sets of each dataset. The representational alignment measurements were also computed on the test sets.

Following Ma et al. (2022), we use different performance metrics for the evaluation of the datasets, as each task has different characteristics. Hence, for Hateful Memes AUROC was reported, while for UPMC-Food101 accuracy was used. For MM-IMDb the F1-Macro score was the main metric, as it is a multi-label classification task with imbalanced classes.

3.2 Implementation

The implementation in this thesis follows the proposed implementation by Lu et al. (2019). For the experiments, a variation of the dual-stream transformer ViLBERT was utilized. Instead of a Faster R-CNN object detector (Ren et al., 2016), a pretrained ViT with 16×16 patches for images of dimension 224×224 was adapted. The image encoder has 12 layers. Following the methodology of the original implementation, we used a 12-layer BERT (Devlin et al., 2019) encoder for the text stream.

Between the text and vision streams, cross-modal attention layers were inserted at different depths of the encoder models. Placement of cross-modal attention modules can be defined by two lists containing the indices of layers the cross-attention is placed before. As seen in Table 3.2, different placements were used for the experiments. Here `text_ids` and `vision_ids` define the indices of the layers before which cross-attention layers are placed in the text and vision streams respectively. As cross-modal attention is bidirectional, the lists are constrained to have the same length. An example for `text_ids = [7]` and `vision_ids = [4]` is presented in Figure 3.2. Here, a text stream before layer 7 (i.e. after layer 6) attends to the latent features of the vision stream before layer 4 (i.e. after layer 3) and vice versa.

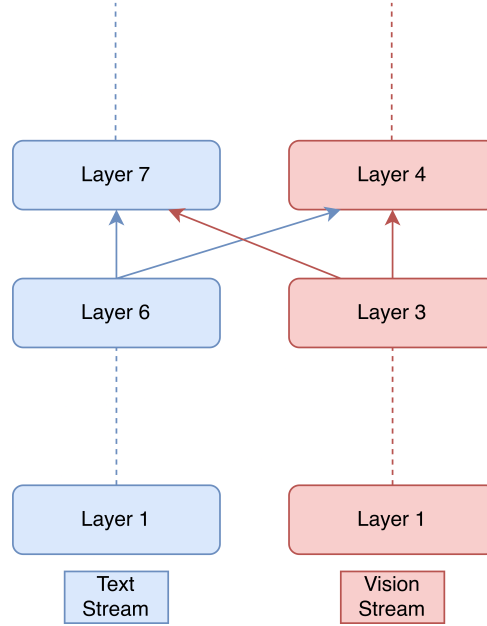


Figure 3.2: Architecture: Cross-attention placed between text-layer 7 and vision-layer 4. Layers without cross-attention perform self-attention

In the original implementation, embeddings were pre-computed using a Faster R-CNN before being passed through a transformer encoder. Therefore, the visual stream had fewer layers than the text stream, requiring asymmetric placements. With the rise of the Vision Transformer (ViT), both streams can have the same depth, allowing symmetric placements. Nonetheless, we experimented with asymmetric architectures, as described in Section 3.3.

For the dual-stream approach, there is no final fused representation, rather two final coordinated representations for each modality (Liang et al., 2023). To facilitate cross-modal capabilities, the separate representations need to be fused. For this thesis, we used the concatenation of both final representations for downstream tasks. Section A.3 covers a minor analysis on the different final fusion methods.

3.3 Model Configurations

To systematically analyze the effect of cross-attention placements, eight different configurations were used for pretraining and fine-tuning. The configurations in combination with their specification are listed in Table 3.2. For comparison, two baselines were tested: A model without any cross-attention layers (Baseline 1) and a model with cross-attention layers at every layer (Baseline 2). For the main architectures, symmetric placements of Early, Middle and Late Fusion were tested, as they are extensively discussed in the literature (Liang et al., 2023). In addition, one asymmetric configuration was tested, to see if asymmetric placements yield different alignment properties. Finally, two optimized configurations were tested, found by hyperparameter optimization using Optuna (Akiba et al., 2019) for Hateful Memes and MM-IMDb respectively. However, the results for this optimization were inconclusive, as discussed in Section A.2.

Name	text_ids	vision_ids
Baseline 1	[]	[]
Baseline 2	[1-12]	[1-12]
Early Fusion	[4,5,6]	[4,5,6]
Middle Fusion	[7,8,9]	[7,8,9]
Late Fusion	[10,11,12]	[10,11,12]
Asymmetric Fusion	[7,8,9,10]	[4,6,8,10]
O-HM	[4,7]	[7,9]
O-MM	[8,10,11,12]	[7,8,10,11]

Table 3.2: Cross-attention placements used for pretraining in this work. O-HM represents the best configuration found by Optuna for Hateful Memes, O-MM for MM-IMDb.

3.4 Training

3.4.1 Pretraining

For the pretraining phase, the three tasks of ViLBERT were adopted. These tasks were inspired by the two tasks used for text pretraining in BERT and adapted to the multimodal setting. While BERT uses Masked Language Modelling (MLM) and Next Sentence Prediction (NSP), ViLBERT extends this to Masked Image Modelling (MIM) and Alignment Prediction (AP).

In MLM, 15% of the tokens in the text stream are masked and the model has to predict the token IDs. The IDs correspond to the total vocabulary size of 30522 (Devlin et al., 2019). For this task, a Cross Entropy loss was used and only applied to the predictions.

In MIM, we used a more sophisticated approach than the original implementation. A contrastive loss was used to optimize the latents of the masked image patches to be similar to the latents of the original unmasked patches, as formalized in Hondru et al. (2025). Although this approach is different from the approach of optimizing MSE loss between original image and the reconstructed patches, Wu et al. (2023) established correlations of the latent representations between the two approaches. In addition, weak and strong data augmentations were utilized, as suggested by Wu et al. (2023). Two data augmentations were applied to all images, resulting in two images I_m, I_o : one strong (horizontal flip, color jitter with brightness=0.3, contrast=0.3, saturation=0.3, hue=0.1, random gaussian blur) and one weakly augmented (minor random rotation, random color jitter and resizing). The strongly augmented image had 10% of patches masked, while the weakly augmented image remained unmasked. Both images were passed through the model $R_o = E_\theta(I_o), R_m = E_\theta(I_m)$, before applying the contrastive loss between

the representations: $L_{MIM} = \text{Contrastive}(R_o, R_m)$. For contrastive loss the InfoNCE loss was used, a prominent loss function in contrastive learning, also used in CLIP (Radford et al., 2021).

$$L_{MIM} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp(\cos(R_m^i, R_o^i)/\tau)}{\sum_{j=1}^N \exp(\cos(R_m^i, R_o^j)/\tau)}. \quad (3.1)$$

τ is the temperature parameter, set to 0.1 as suggested in Radford et al. (2021).

In AP, the task objective is to predict whether a given text-image pair (T_i, I_i) is aligned. For this task, pairs in the Conceptual Captions dataset were assumed aligned. As mentioned above, there are artifacts in the dataset due to the nature of the data collection step. Formally, this task can be described by a function $\phi : I \times T \rightarrow \{0, 1\}$ that predicts if a given image I and text T match: $\sigma(\phi(I_i, T_i)) > 0.5 \Rightarrow \text{aligned}$. For this task, a BCE loss was used.

During implementation, we tested whether all three tasks are necessary to achieve good alignment, as MLM and MIM are proxy tasks and a purely AP-driven pretraining seemed more appropriate. However, smaller experiments showed that AP alone failed to converge. Details are discussed in Section A.3.

3.4.2 Fine-tuning

After pretraining different cross-attention placements for 7 epochs with a learning rate of 10^{-4} , the models were fine-tuned on the three downstream tasks described above. To mitigate variance, training was performed using three different seeds and the mean and standard deviation were reported across the runs. Each task used a task-specific classification head consisting of a small MLP with a hidden dimension of 1024 and ReLU activation. The MLP maps the concatenated representations to the number of classes in each task. For both pretraining and fine-tuning, the AdamW optimizer was used with a cosine learning rate scheduler (Loshchilov and Hutter, 2017) with linear warmup for the first 10% of the total training steps.

Models were trained for 15 epochs to ensure convergence across all architectures. Early stopping was performed using validation accuracy across all tasks for consistency, although task-specific metrics (AUROC for Hateful Memes, F1-Macro for MM-IMDb) were used for final evaluation.

3.5 Representational Alignment Analysis

This thesis includes three main experiments regarding representational alignment metrics. First, evolution of alignment across different depths is analyzed. Second, correlation analysis between alignment metrics and downstream task performance is conducted. Third, layer-to-layer comparisons across all layers are performed to analyze asymmetric cross-layer alignment patterns.

Both modality streams use pretrained encoders with a default depth of 12. By inserting cross-modal attention between the two modalities, the depth increases. The number of cross-attention layers varied across configurations. A consistent approach was adopted that allowed comparison across different architectures.

To analyze representational alignment metrics at every layer, we performed several forward passes on the test set, collecting 1024 samples of hidden representations. As there are two streams, for each input pair $(\text{text}, \text{image})$, we collected the hidden representations of both streams at each layer depth. Ensuring comparability between different placements (e.g. 2 cross-attentions vs. 5 cross-attentions), representations from the original 12 layers were collected.

Storing all token representations requires substantial memory. One sample requires approximately $197 \times 768 \times 12 \times 4 \approx 7.3$ MB for vision and language tokens, totaling 14.6 MB per sample.

For 1,024 samples, this amounts to approximately 14.7 GB, excluding intermediate memory for metric computation.

Following Huh et al. (2024); Tjandrasuwita et al. (2025), we extracted only CLS tokens from each layer, reducing memory requirements to approximately 74 MB for 1,024 samples while preserving global layer representations. Stability analysis (Appendix A.1.2) confirmed reliable results with sample sizes above 512; we used 1,024 samples to match prior work. These samples were drawn from the test set. Before the similarity metrics were calculated, CLS tokens of both modalities were normalized by the L^2 norm, following Huh et al. (2024).

For the analysis, four complementary metrics were computed:

- CKA: Global alignment based on structural similarity.
- mkNN: Local alignment based on nearest neighbors.
- SVCCA: Dimensionality reduction in combination with correlation.
- Orthogonal Procrustes: Geometric alignment via orthogonal transformation.

For cross-modality alignment, these metrics were computed on the same layer depth of both modalities. For example, the CKA score at layer 5 is computed between the CLS tokens of the text and vision streams after layer 5. This approach enables a layer-wise analysis of the evolution of alignment within the two streams.

Correlation Analysis

In addition to the layer-wise analysis, a correlation analysis between alignment metrics and downstream task performance was conducted. This involved calculating Pearson and Spearman correlation coefficients between alignment scores at the final layer and task-specific performance metrics (AUROC for Hateful Memes, F1-Macro for MM-IMDb, Accuracy for UPMC-Food101). We focus on final-layer alignment as it most directly influences downstream task behavior. This analysis aimed to uncover potential relationships between representational alignment and model effectiveness on multimodal tasks. For this analysis, all 72 models were evaluated: 8 architectural configurations $\cdot$ 3 tasks $\cdot$ 3 seeds. We collected task performance metrics from the test set and alignment values (mkNN, CKA, SVCCA, OP) at the final layer. To validate robustness given non-independence, we additionally computed correlations using the mean alignment and performance across seeds per configuration, confirming similar correlation patterns.

3.6 Linear Probing

Linear probing is a common method to quantify representation learning capabilities of models (Radford et al., 2021). For a trained model, linear layers are trained on different tasks, showing generality of a pretrained model. To analyze layer-wise behavior of multimodal models, linear probing was utilized in this work. For each fine-tuned model, linear layers were fitted on top of each layer’s latents. This provides insights into how well each layer’s representations support downstream tasks.

For each layer $n \in [1, \dots, 12]$ under investigation, vision and language representations (CLS tokens) were extracted at layer n . The modalities were concatenated into a single representation: $\mathbf{h}_n = [\mathbf{R}_n^{(v)}; \mathbf{R}_n^{(l)}]$ with dimensionality of $2 \cdot d = 1536$. Subsequently, a linear projection was learned for the downstream task. The model was frozen and only the parameters for the linear layer were trained. For this experiment, a learning rate of 10^{-3} with AdamW optimizer (Loshchilov and Hutter, 2019) was used to train the linear layer for 3 epochs.

In addition, representational alignment metrics (RAM) were computed at each layer to correlate representational alignment with task performance.

Chapter 4

Results

4.1 Task Performance

Results show Late Fusion consistently outperforms other cross-attention placements. Table 4.1 reports performance metrics across all tasks. Each stream benefits from longer unimodal processing before correspondence between the two streams. This allows each modality to develop rich, task-specific representations prior to interaction. Interestingly, the fine-tuning-optimized cross-attention placements (O-HM and O-MM) do not outperform the Late Fusion strategy, suggesting that pretraining has different characteristics from fine-tuning-only approaches. These optimized configurations yield similar performance across all tasks, especially in Hateful Memes and MM-IMDb.

Model	HM AUROC	MM-IMDb F1	UPMC-Food101 Accuracy
Baseline 1	0.6542 ± 0.0016	0.4825 ± 0.0084	0.8871 ± 0.0012
Early Fusion	0.7226 ± 0.0034	0.5118 ± 0.0206	0.8930 ± 0.0001
Middle Fusion	0.7497 ± 0.0090	0.5334 ± 0.0075	0.9181 ± 0.0013
Late Fusion	0.7637 ± 0.0041	0.5448 ± 0.0120	0.9278 ± 0.0005
Asymmetric Fusion	0.7245 ± 0.0018	0.5274 ± 0.0050	0.9010 ± 0.0002
O-HM	0.7460 ± 0.0025	0.5339 ± 0.0081	0.9178 ± 0.0004
O-MM	0.7401 ± 0.0050	0.5340 ± 0.0040	0.9215 ± 0.0010
Baseline 2	0.7074 ± 0.0013	0.5283 ± 0.0036	0.9094 ± 0.0010

Table 4.1: Results across all datasets using task-specific primary metrics: AUROC for Hateful Memes (HM), F1-Macro Score for MM-IMDb, and Accuracy for UPMC-Food101. $n=3$ seeds per configuration. Baseline 1 has no cross-attention layers, Baseline 2 has cross-attention at every layer. O-HM represents the best configuration found by Optuna for Hateful Memes, O-MM for MM-IMDb.

Experiments show task-dependent correlations between performance and representational alignment metrics (RAM). The results are presented in Table 4.2. A high correlation between performance and RAM for UPMC-Food101 is notable, where all metrics show strong correlations (Pearson and Spearman) with performance. In general, the correlation between Procrustes and performance is negative, which is expected, as OP is a dissimilarity metric. OP notably shows statistical significance in both UPMC-Food101 and MM-IMDb with strong correlations of -0.915 and -0.813 (Pearson).

For MM-IMDb, Pearson correlations are higher with greater statistical significance compared to Spearman. This suggests a strong linear relationship. Spearman, being rank-based, is more robust to outliers (Schober et al., 2018) and shows weaker correlations, indicating the relationship

may not be monotonic across all data points. However, as noted in Schober et al. (2018), when Pearson and Spearman differ significantly, it indicates potential violations of the underlying assumptions.

In Hateful Memes, a significant divergence between both correlation metrics is observed. Therefore, these values should be interpreted with caution. Furthermore, Hateful Memes exhibits highly unstable correlations, with only SVCCA showing a statistically significant Pearson correlation of 0.709 ($p=0.049$).

Task	Metric	Pearson ρ	Pearson p	Spearman ρ	Spearman p
Hateful Memes	mkNN	+0.634	0.091	+0.381	0.352
	CKA	+0.456	0.256	+0.071	0.867
	SVCCA	+0.709	0.049	+0.405	0.320
	OP	-0.075	0.860	+0.071	0.867
MM-IMDb	mkNN	+0.867	0.005	+0.524	0.183
	CKA	+0.916	0.001	+0.762	0.028
	SVCCA	+0.914	0.001	+0.619	0.102
	OP	-0.915	0.001	-0.881	0.004
UPMC-Food101	mkNN	+0.766	0.027	+0.857	0.007
	CKA	+0.908	0.002	+0.929	0.001
	SVCCA	+0.939	0.001	+0.976	0.000
	OP	-0.813	0.014	-0.905	0.002

Table 4.2: Correlation between alignment metrics and performance across tasks. $n = 8$ per task

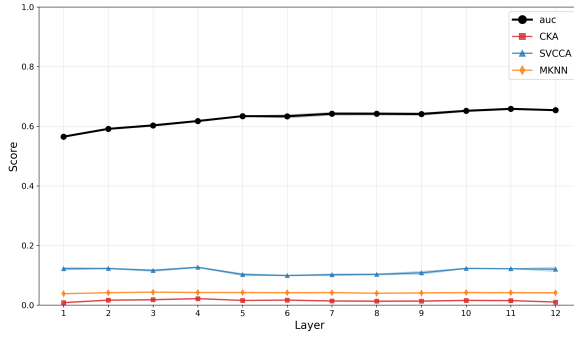
The number of models per task is limited to 8 configurations, which limits statistical power. Correlation analysis on the pooled results ($n=24$: 8 models $\times$ 3 seeds) resulted in similar correlations and statistical significance, confirming robustness of the findings.

Orthogonal Procrustes shows fundamentally different correlation properties compared to other metrics. While CKA, SVCCA, and mkNN show visually interpretable layer-wise trajectories, Procrustes exhibits noisy layer-wise patterns that nonetheless correlate strongly with task performance (MM-IMDb: $\rho = -0.915, p = 0.001$; UPMC-Food101: $\rho = -0.813, p = 0.014$). The inverse correlation, higher Procrustes distance associated with lower performance, indicates that the geometric divergence between modalities matters for downstream tasks. As OP is a dissimilarity measure, the negative correlation can be interpreted as: lower distance between final representations leads to better performance. However, this relationship is not captured in layer-by-layer analysis. We hypothesize that representations with better performance are already embedded in similar geometric spaces. This connects indirectly to the Platonic Hypothesis. These distinguishable results highlight the importance of a diverse metric selection.

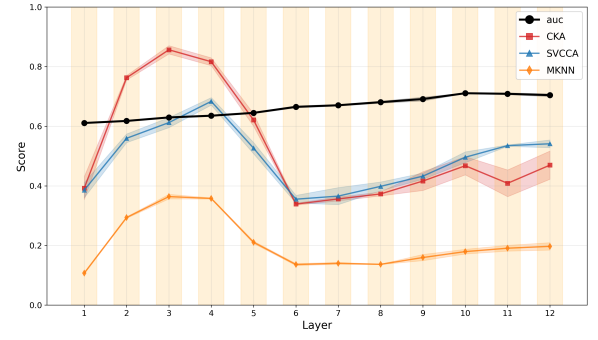
4.2 Layer-wise Probing Results

Layer-wise linear probing provides rich insights into the inner workings of alignment in the dual-stream architecture. There are clear differences between the two baselines, which form the extrema of a spectrum. While Baseline 1 has no cross-attention placements, Baseline 2 incorporates symmetric cross-attention at every layer. Layer-wise performances plotted against RAM are shown in Figures 4.1, 4.2, and 4.3.

Baselines: For Baseline 1, representational alignment slightly increases with depth of the

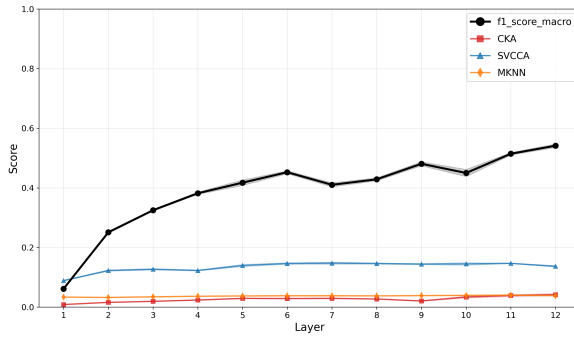


(a) Baseline

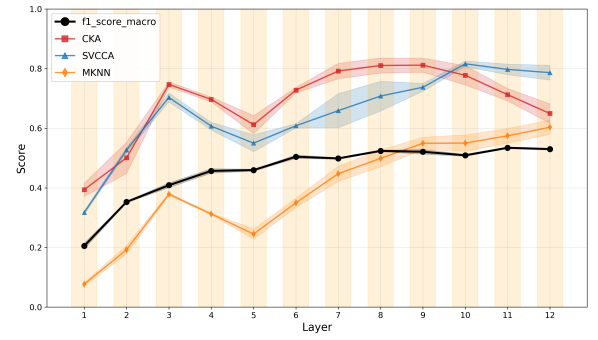


(b) Full cross-attention

Figure 4.1: Hateful Memes: AUROC vs CKA, SVCCA and mkNN

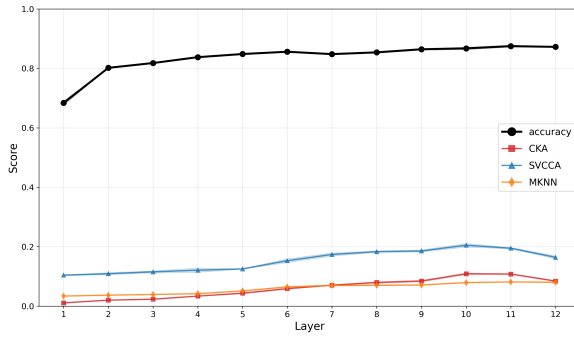


(a) Baseline

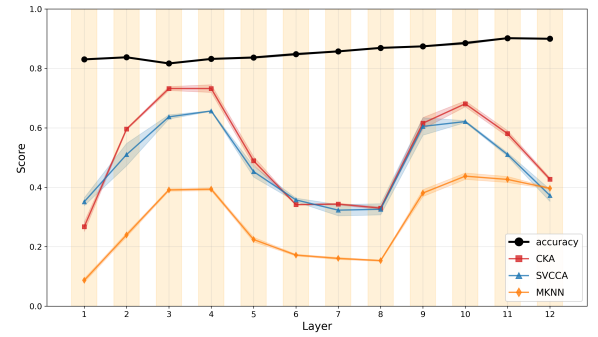


(b) Full cross-attention

Figure 4.2: MM-IMDb: F1-Macro vs CKA, SVCCA and mkNN



(a) Baseline



(b) Full cross-attention

Figure 4.3: UPMC-Food101: Accuracy vs CKA, SVCCA and mkNN

transformer. Furthermore, this effect is prominent in MM-IMDb and UPMC-Food101, the datasets with more similarity between modalities. These findings align with Tjandrasuwita et al. (2025) that tasks with redundant modalities exhibit higher representational alignment. This supports the hypothesis of unimodal encoders generalizing to similar learned representations, proposed by Huh et al. (2024). Baseline 2 shows a pattern of two peaks in representational alignment across all datasets, as visible in b) of Figures 4.1, 4.2, and 4.3. An initial peak occurs around layers 4-5, followed by a dip and a second peak around layers 9-11. To test whether both peaks could be leveraged, we constructed hybrid architectures combining early and late placements. Results indicate this is not possible; hybrid models consistently underperform pure Late Fusion (e.g., 0.7585 vs. 0.7637 on Hateful Memes). This suggests that early cross-attention forces a premature collapse of the representation space, creating a bottleneck that persists even with subsequent refinement layers, thereby negating the benefits of Late Fusion.

Cross-Attention increases Alignment

Analysis shows that cross-attention layers strongly increase representational alignment. A t-test compared metrics at consecutive layers for cross-attention versus non-cross-attention placements. It revealed a significant increase in representational alignment for CKA, SVCCA and mkNN. Notably, the first cross-attention layer in the model typically increases alignment the most, while the subsequent placements exhibit substantial volatility. This suggests that major representational restructuring occurs within the cross-attention layers. The later in the stream cross-attention is placed, the greater the increment in representational alignment. This suggests that later cross-attention layers have a stronger impact on aligning representations, possibly because the unimodal streams develop richer features before interaction. Figure 4.4 compares Early- versus Late Fusion in Hateful Memes, showing that Late Fusion has a stronger alignment increase after cross-attention layers.

The improvement of alignment is prominent in all tasks; Table 4.3 reports the values for Hateful Memes. For results on MM-IMDb and UPMC-Food101, see Appendix B.1.

Metric	Mean (No CA)	Mean (CA)	Δ (No CA)	Δ (CA)
CKA	0.079	0.301	0.014	0.181***
mkNN	0.062	0.158	0.003	0.066***
SVCCA	0.189	0.419	0.016	0.158***
OP	13.38	22.41	1.74	5.12***
Performance	0.646	0.720	0.011	0.031***

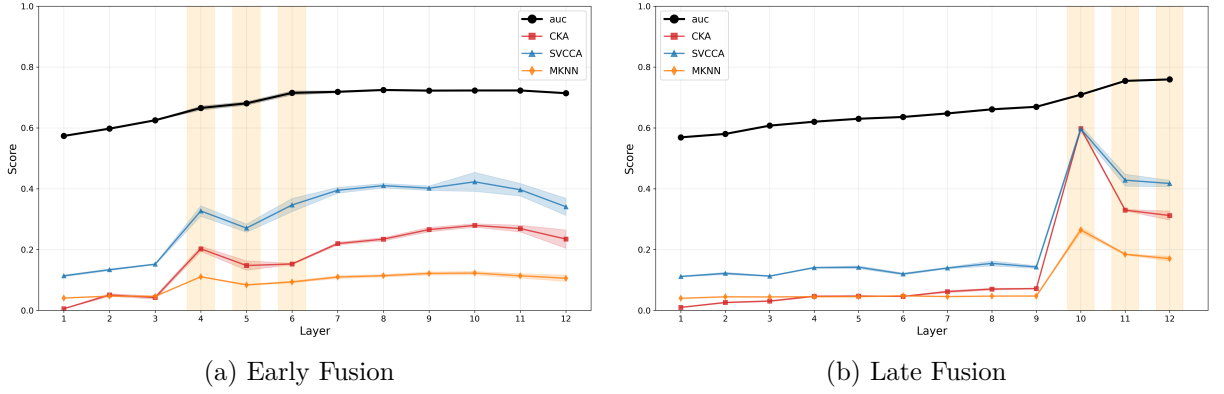
CA = Cross-Attention; Δ = mean consecutive layer difference; n=117 (No CA), n=27 (CA)

***p < 0.001, Performance = AUROC.

Table 4.3: Effect of cross-attention on alignment metrics in Hateful Memes. Cross-attention significantly increases all alignment metrics.

Notably, layer-wise performance differences between cross-attention and non-cross-attention placements reveal that **performance on Hateful Memes increases significantly following the introduction of cross-attention layers** (p<0.05). For this task, cross-attention layers show a 3× improvement in performance. Conversely, MM-IMDb and UPMC-Food101 exhibit the opposite effect, where performance does not increase significantly after cross-attention placements. This finding demonstrates that for complicated tasks like Hateful Memes, cross-modal interactions allow for better task-specific learning. For MM-IMDb, we observe the opposite; for UPMC-Food101, differences were not statistically significant.

Inter-model similarity analysis reveals that alignment emerges predominantly after the in-

Figure 4.4: **Hateful Memes**: Early vs. Late Fusion

roduction of cross-attention modules. As shown in Figure 4.5, high cross-layer CKA scores appear mainly in layers 10-12, while earlier layers exhibit almost no cross-modal similarity. This pattern suggests that both streams initially learn modality-specific transformations, extracting features independently. Only after sufficient unimodal processing the cross-attention layers facilitate meaningful interactions, aligning representations in the deeper layers. The resulting high similarity in later layers reflects semantic alignment rather than low-level coupling.

In contrast, the Early Fusion configuration demonstrates alignment emerging immediately at cross-attention layers 4-6. Representations from both streams are coupled after fusion and show noisy patterns in later layers. Notably, the visual encoder shows uniformly high intra-stream CKA between layers 6-10, suggesting minimal representational change beyond this point. Such high self-similarity can be interpreted as representation oversmoothing.

Optimization for CKA does not increase Performance

To test whether the observed correlation between CKA and performance is causal, models with an additional CKA loss term were trained optimizing for structural similarity:

$$\mathcal{L}_{total} = \mathcal{L}_{tasks} + \lambda \cdot \text{CKA}(\text{text}, \text{vision}) \quad (4.1)$$

where $\lambda = 0.2$.

Table 4.4 shows results for both Early and Late Fusion configurations. Optimizing for CKA directly yields no performance improvement in Late Fusion and causes pretraining failure in Early Fusion. Here, the Alignment Prediction task collapses (Figure 4.6)

Model	Accuracy	F1-Macro Score
Late Fusion	0.9278 $\pm$ 0.0005	0.9272 $\pm$ 0.0006
Late Fusion (CKA)	0.9262 $\pm$ 0.0015	0.9256 $\pm$ 0.0016
Early Fusion	0.8930 $\pm$ 0.0001	0.8921 $\pm$ 0.0001
Early Fusion (CKA)	0.8566 $\pm$ 0.0018	0.8569 $\pm$ 0.0019

Table 4.4: Results on UPMC-Food101 dataset.

Pretraining on Early Fusion configuration with CKA optimization term added was mostly unsuccessful, as reported in Figure 4.6. We hypothesize that optimizing directly for structural similarity is not a sufficient objective. Two representations can be structurally similar but contain little discriminative complexity. Davari et al. (2022) demonstrated ways to manipulate

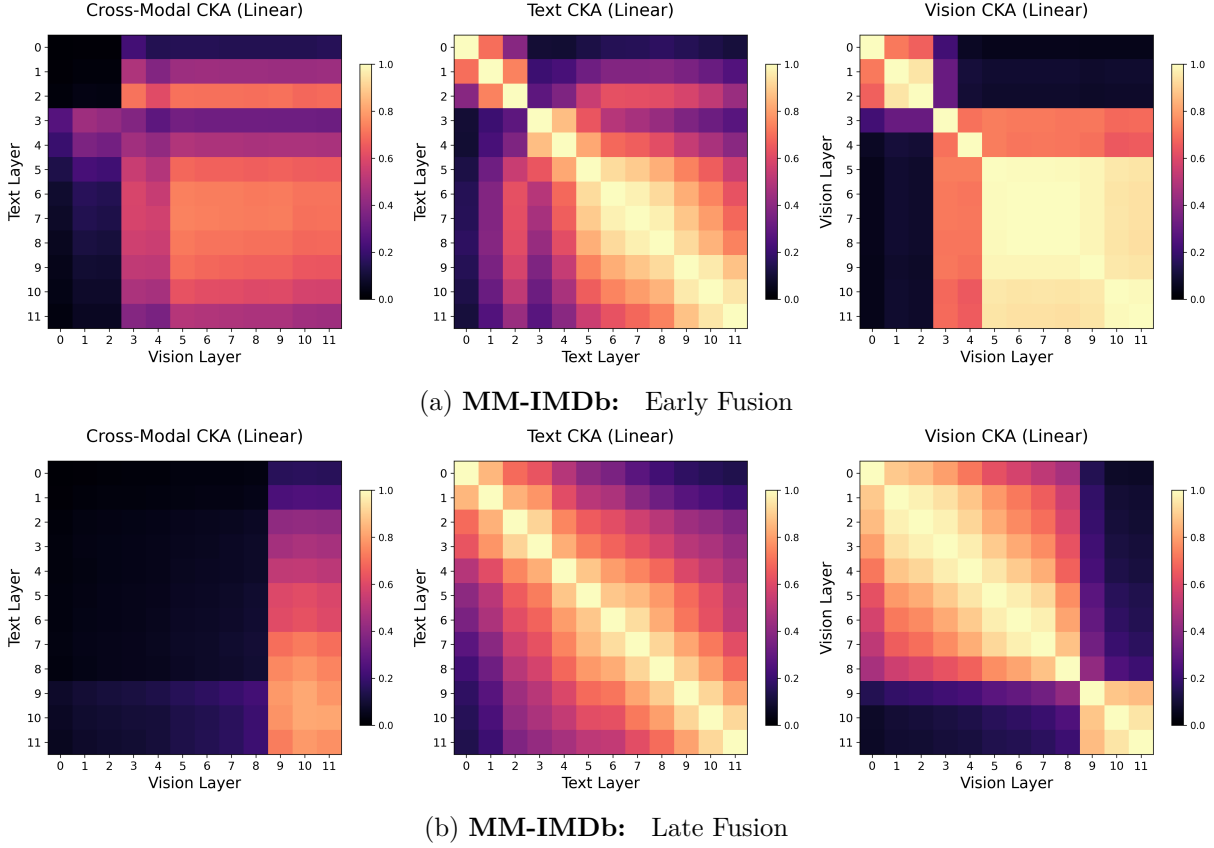


Figure 4.5: Layer-to-layer CKA matrices for Early Fusion vs Late Fusion in MM-IMDb. Late Fusion shows clear alignment in later layers in cross-modal representations, while Early Fusion shows noisy patterns

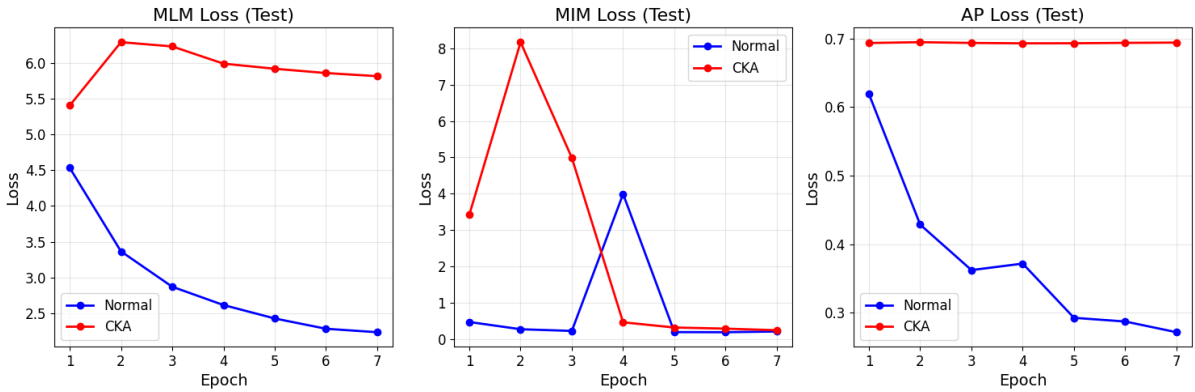


Figure 4.6: Pretraining losses of Early Fusion (Normal) vs Early Fusion CKA optimized (CKA): Masked Language Modelling (MLM) shows significant degradation. Masked image prediction (MIM) sufficiently learns and converges to a similar loss level as normal early fusion. Alignment Prediction (AP) completely fails to learn.

CKA values without affecting performance, suggesting the model implicitly learns to manipulate the metric. Hence, CKA is not suitable as a direct optimization objective.

We hypothesize that the added loss term conflates the actual loss and destabilizes learning. This indicates that the loss landscape of optimizing for CKA is either orthogonal to the task’s terms or rather simple to learn with clear extrema, obfuscating the pretraining task’s losses.

4.3 Mechanistic analysis: Why Late Fusion outperforms

To understand why Late Fusion outperforms other placements, two complementary metrics were used for analyzing representational evolution across layers: intra-modal CKA similarity between consecutive layers and effective dimensionality.

Cross-attention Restructures Representations

For any latent representation $\mathbf{R}_i$ at layer l_i similarity with the consecutive layer l_{i+1} was computed using $\text{CKA}(\mathbf{R}_i, \mathbf{R}_{i+1})$. The similarity was computed for each stream respectively. This provides insights into the magnitude of transformations happening to self- and cross-attention layers, as well as whether the trajectory consists of balanced transformations or sudden jumps. Figure 4.7 reveals that cross-attention placements (in gray) induce abrupt representational transformations. While layers before cross-interactions show gradual changes (high CKA similarity, > 0.8), cross-attention layers introduce significant restructuring (low CKA similarity, 0.2-0.4). This indicates that cross-modal attention does not only add information but fundamentally reorganizes the representation space.

Critically, **Early Fusion undergoes the largest structural changes at a formative stage of learning**. By placing cross-attention at layers 4-6, Early Fusion impedes modality specific feature development before representations mature. This early disruption limits the richness of features that can develop. This can be further seen in Section 4.3. As the subsequent layers exhibit intra-modality CKA values spiking to ≈ 1.0 , there are no further major transformations recovering from it.

Notably, vision is more susceptible to structural changes than text. Across all fusion strategies, vision shows stronger restructuring at cross-attention layers than text. This suggests vision representations require more adaptation to align with text, while text representations remain relatively stable.

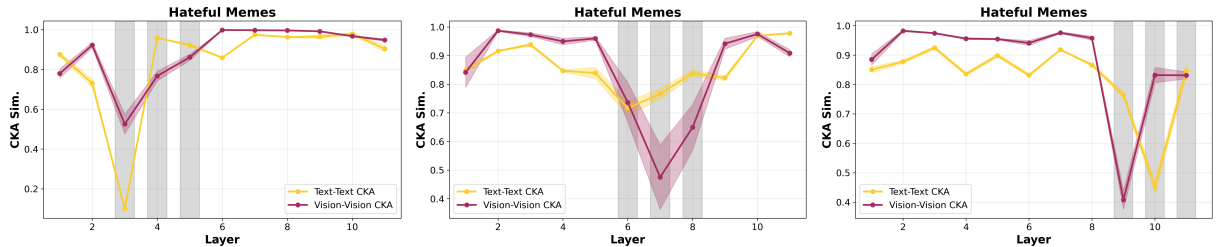


Figure 4.7: Intra-modal CKA evolution across layers for Early Fusion (left), Middle Fusion (middle) and Late Fusion (right). Each plot shows the CKA trajectory for text (yellow) and vision (purple) separately. The y-axis represents similarity between two consecutive representations $\text{CKA}(\mathbf{R}_i, \mathbf{R}_{i+1})$. The x-axis represents the layer transition.

Early Fusion Constrains Development

Figure 4.8 reveals an even more striking pattern. For text and vision separately, the representation is reduced to n dimensions using PCA, where n is the number of dimensions that explain 95% of the variance. Here, the threshold variance is fixed, while the number of dimensions is the variable of interest. For each architecture we plot the trajectory of the number of dimensions across layers.

Text representations (yellow) develop steadily to 250-300 dimensions across all fusion strategies, indicating robust and independent feature learning. Vision representations (purple), however, exhibit a fundamental fragility. As seen in the architectural baselines (see Appendix B.5), the vision encoder requires cross-modal interaction to expand its representational capacity. Without it, it remains low-dimensional. While dimensionality fluctuates in subsequent layers, it remains substantially constrained compared to Late Fusion. In Hateful Memes (4.8), vision remains suppressed below 30 dimensions; in MM-IMDb and UPMC-Food101 (Figure B.4) partial recovery occurs but dimensionality stays well below the 150-200 dimensions achieved in Late Fusion.

In contrast, Late Fusion allows vision to develop rich representations (150-200 dimensions) before fusion. Both modalities undergo controlled dimensionality reduction only at the fusion point (layer transitions 9–11), after modality-specific features are extracted. Note that the final effective dimension for a task is quite similar across configurations (except in Hateful Memes).

These findings correspond to the layer-wise comparison in Figure 4.5 and are shown in more detail in Appendix B.2. The vision-to-vision figures for Early Fusion show highly similar latent features across layers 6-12.

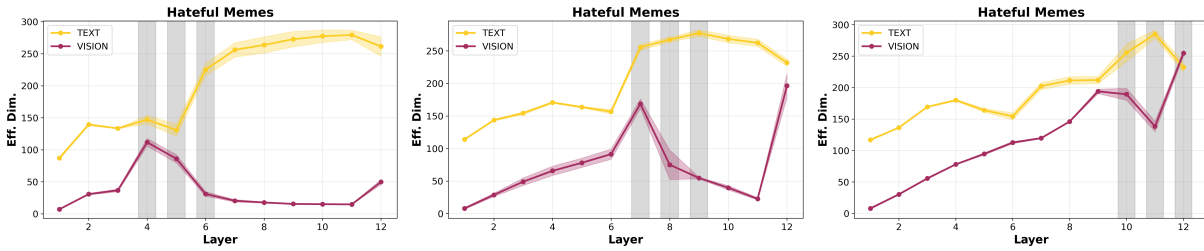


Figure 4.8: Effective Dimensionality evolution across layers for Early Fusion (left), Middle Fusion (middle) and Late Fusion (right). Each plot shows the effective dimensionality trajectory for text (yellow) and vision (purple) separately. The y-axis represents the effective dimensionality. Note the **dimensionality collapse** in the Vision stream (purple) for Early Fusion compared to the preserved richness in Late Fusion.

Collapse Hypothesis

We hypothesize that the observed dimensionality collapse is induced by the fundamental difference in the abstraction level of the input embeddings. While text embeddings are discrete and token-wise, visual embeddings are simply 16×16 pixel patches. By splitting the image into patches, semantic information is fragmented. Patches contain low-level signals instead of high level semantics.

As illustrated in Figure 4.9, this results in a fragmented representation for vision. A majority of the image patches contain only background texture or noise (Figure 4.9a). In contrast, text tokens carry specific semantic meaning immediately (Figure 4.9b). The ViT model requires several layers of self-attention to extract the high-level information. Semantic signals embedded in the high entropy patches are extracted in the first layers. This is directly related to the

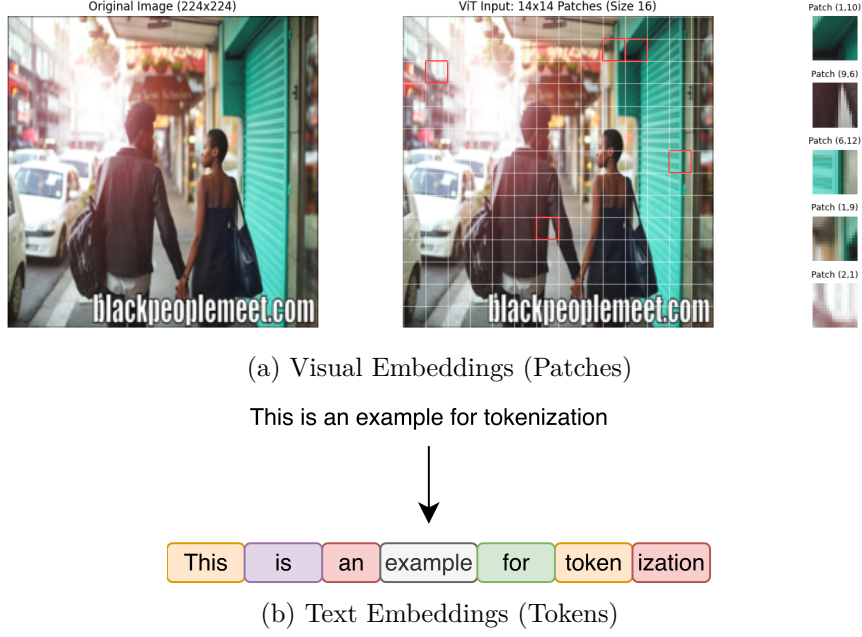


Figure 4.9: Contrast in abstraction level. (a) A single visual patch contains low-level pixel information with high noise. (b) A single text token conveys a distinct semantic concept.

notion of raw and abstract modalities. While raw image patches lack semantic invariance, encoded representations are more abstract.

In Early Fusion, cross-modal attention forces these raw, low-level patches to align with high-level text tokens at layers 4–6. We argue that the vision stream fails to extract high-level features at this stage. Forced to minimize loss against semantically high-level text tokens, the vision stream collapses into a low-dimensional subspace. Late Fusion avoids this by allowing the vision stream to mature (layers 1–9) before interaction occurs.

As shown in Figure B.5 in the Appendix, the vision stream naturally resides on a low-dimensional manifold (Goodfellow et al., 2016). This is evidenced by Baseline 1 (no cross-modal interaction), where the vision stream maintains an effective dimensionality below 70, compared to the text stream’s 150. Only through the interaction of both modalities does the dimensionality exceed the baselines. We hypothesize that the underlying multimodal manifold has higher dimensionality than each modality alone, and that optimal cross-attention placement is required to learn this manifold without premature collapse.

Implications: These findings reveal that vision is fragile. While the text stream maintains robustness, Early fusion forces the vision stream early on to compress to a low-dimensional subspace. In contrast, the other configurations allow vision to develop rich, high-dimensional features before fusion. Late Fusion preserves both modality-specific information ($I(R_V)$, $I(R_T)$) and cross-modal information ($I(R_V; R_T)$). We hypothesize that this explains its 2-5% performance advantage.

4.4 Discussion

This work demonstrates that Late Fusion consistently outperforms Early- and Middle Fusion across all tested datasets, achieving 2-5% performance gains over alternative placements. This finding is architecture-dependent and differs from prior work on other multimodal architectures.

Experiments by Ma et al. (2022) demonstrated that Early Fusion models performed best,

while our experiments show Late Fusion consistently outperforming other cross-attention placements across all datasets. Ma et al. (2022) utilizes a ViLT (Kim et al., 2021) architecture. Conversely, Nagrani et al. (2021) experimented with vision and audio transformers, resulting in middle fusion performing best. They used a custom fusion strategy named Multimodal Bottleneck Transformer. Our findings diverge from theirs and show that the optimal cross-attention placement is architecture-dependent. This implies that the distinction between Early- and Late Fusion is not equally applicable across all architectures and modalities.

While Guo et al. (2023)’s experiments using a dynamic cross-attention placement strategy resulted in task-specific placements to handle modality misalignment, our findings offer a nuanced perspective. We found that static Early Fusion configurations induce a severe dimensionality collapse in the vision stream, where effective dimensionality drops from ~ 150 to $\sim 10 - 20$ dimensions (an 87% reduction) immediately following cross-attention. Guo et al. (2023) links misalignment to discrepancies between input modalities; our mechanistic analysis confirms this, showing that forcing raw ViT patches to align early with text constrains the representational capacity of the vision encoder. Thus, while Switch-BERT proposes dynamic routing, our results suggest that for ViT-based dual-stream models, simply delaying fusion to prevent this representational collapse is a robust solution.

Our experiments showed a clear correlation between performance and representational alignment metrics. This correlation is task dependent and most promising for UPMC-Food101. Interestingly, the highest correlation is observed for the simpler task UPMC-Food101 and the moderately simple task of MM-IMDb. For Hateful Memes, the hard task that requires complex multimodal reasoning, the correlations are weak and unstable. This finding aligns with Tjandrasuwita et al. (2025)’s work. They reported a high correlation between alignment metrics and performance for tasks with redundant modalities. Conversely, tasks with high uniqueness in each modality have low and unstable correlation.

The field of representational alignment metrics is still new with many open questions. The high-dimensional spaces in transformers offer a high degree of freedom. It involves trade-offs reducing high-dimensional representations to a scalar metric bounded by $[0, 1]$. While metrics like SVCCA and CKA are useful, they do not provide a complete picture of representational similarity. As shown by Davari et al. (2022), CKA can easily be manipulated to different values without affecting performance. This motivated our use of four complementary metrics (CKA, SVCCA, mkNN, OP) with different invariance properties, for a more robust analysis.

4.4.1 Local versus Global

A limiting factor in this analysis is the scope of local alignment. We employed local metrics like mkNN, which assess neighborhood structure in the representation space across 1024 samples ($1024 \text{ samples} \times 768 \text{ dimensions}$). However, this captures local alignment at the representation level, not at the token level. True token-level local alignment would require comparing individual text tokens to their corresponding image patches, measuring fine-grained concept correspondence. For example, in vision-language tasks, specific text tokens should align with particular image patches representing the same concept (Liang et al., 2023). Computing such pairwise token alignments would require substantially more memory and computation. This remains an interesting direction for future work, particularly since recent literature has focused primarily on global alignment (CLS Token) (Huh et al., 2024; Tjandrasuwita et al., 2025; Klabunde et al., 2023).

Figure 4.10 illustrates an example from the pretraining set of Conceptual Captions (Sharma et al. (2018)). The caption is "a commuter train travels next". Because only a small portion of the image contains the actual train, only a few of the 196 patches depict it. The upper



Figure 4.10: Sample from conceptual captions with caption: "a commuter train travels next" (Sharma et al., 2018).

quarter has no train at all. This illustrates why token-level alignment is important but was computationally infeasible in this work.

4.5 Revisiting Research Questions

We now explicitly address the research questions posed in the introduction:

RQ1: How does cross-attention placement influence the evolution of representational alignment across layers? Cross-attention blocks induce a direct, statistically significant increase in alignment metrics (CKA, SVCCA), with the initial block typically inducing the largest shift. Unlike the gradual transformations of self-attention, cross-attention introduces abrupt representational restructuring, as evidenced by low CKA similarity between consecutive layers.

RQ2: What is the relationship between representational alignment metrics and downstream task performance? Correlation is context-dependent. We observed strong correlations ($\rho > 0.9$) for tasks with redundant modalities (e.g., UPMC-Food101) but weak, unstable correlations ($\rho \approx 0.7$) for complex tasks (e.g., Hateful Memes). This suggests that high geometric alignment is not a universal proxy for performance, particularly when tasks require complementary cross-modal reasoning rather than simple overlap.

RQ3: What representational changes occur when cross-attention is introduced at different depths? The most critical change is a *dimensionality collapse* in the vision stream. Introducing cross-attention early (layers 4-6) causes the effective dimensionality of visual representations to drop by $\sim 87\%$ (from ~ 150 to ~ 20), permanently constraining model capacity. In contrast, Late Fusion preserves high dimensionality (150-200) deeper into the network, allowing for rich unimodal feature extraction before integration.

RQ4: How do these alignment patterns vary across tasks with different modality characteristics? While Late Fusion is superior, the performance gap widens for complex tasks requiring detailed feature extraction. We observed a consistent asymmetry across all tasks: the vision stream undergoes significantly more severe structural restructuring than the text stream. This indicates that ViT-based visual representations are inherently more fragile and susceptible to interference from early cross-modal interaction.

Chapter 5

Conclusion

This thesis systematically analyzed representational alignment in dual-stream V&L transformers across eight different cross-attention placements and three multimodal datasets. Each configuration underwent pretraining and fine-tuning. To quantify alignment, representational alignment metrics (CKA, SVCCA, mkNN, and Orthogonal Procrustes) were utilized. These metrics measured similarity between latent representations in the network. We analyzed intra-modality and inter-modality alignment across layers. In addition, we performed layer-wise linear probing on fine-tuned models to assess task performance in relation to alignment.

Experiments across multimodal tasks demonstrated that Late Fusion consistently outperformed Early and Middle Fusion configurations. Representational alignment metrics correlated with performance, the strength of this relationship depended on task characteristics: strong correlations emerged for redundant-modality tasks like UPMC-Food101 ($\rho = 0.939$), while complex tasks like Hateful Memes showed weaker correlations ($\rho = 0.709$). Furthermore, directly optimizing for CKA did not improve performance, but rather hindered learning in some configurations.

It was shown that cross-modal interactions increased similarity of representations in both modalities. Cross-attention learned to transform both streams and coordinate them into similar structures. Furthermore, results showed that cross-attention not only added information from one modality, but fundamentally restructured the representation space. These findings suggest that cross-attention fuses distinct modalities together and creates new representations with a distinguishable spatial structure.

Further investigations into the mechanistic differences between the three fusion strategies revealed severe dimensionality reduction induced by cross-attention layers. While Late Fusion strategies allow for higher dimensional feature extraction before fusion, Early Fusion lacks this stage. Early Fusion caused 85-90% dimensionality collapse in vision representations that persisted through subsequent layers, constraining the feature richness available for downstream tasks.

In addition, experiments revealed an asymmetry for vision: vision representations were more strongly affected by cross-attention, undergoing larger structural changes and dimensionality reductions than text across all fusion strategies. Early Fusion constrained vision’s effective dimensionality drastically, limiting feature richness. In contrast, Late Fusion preserved modality-specific information before integration, explaining its superior performance.

Limitations and Future Work

While this thesis provides novel insights into representational alignment in multimodal dual-stream architectures, there is a limitation. For the representational space across a dataset, only

global alignment (CLS token) was analyzed. Future work should investigate local alignment metrics to complement global analyses.

Moreover, the findings are limited to the specific architecture and datasets used. Different architectures may exhibit different fusion dynamics, as shown by prior work finding early fusion optimal in ViLT or middle fusion optimal in audio-vision transformers. Future research should explore other architectures, modalities, and larger-scale datasets to validate generalizability.

AI Disclosure

For this work, Grammarly was used for spelling and grammar correction. GitHub Copilot was used to assist with routine coding tasks and plot generation. No AI tools were used for idea generation, experimental design, analysis, or writing the thesis text. The ideas, content, analyses and results are my work.

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Appendix A

Supplementary Experiments and Analysis

A.1 Robustness Analysis

To reduce the complexity of the experiments, stability of metrics with smaller sizes was tested. Here, we utilized Spearman Correlation to show that the representational alignment metrics have a similar ordering with different sample sizes.

A.1.1 Dataset size

This thesis computes representational alignment metrics (RAM) for every layer in every model. As this work’s analysis includes eight pretrained models, where each pretrained model is fine-tuned on three tasks with three seeds and for each model 12 layers are analyzed in every epoch, the number of computations is high. To minimize computational load, we determined an optimal sub-sample size via correlation analysis. In previous works (Huh et al., 2024), a correlation analysis between RAM was conducted, but did not focus on impact of dataset size on RAM.

For this work, correlation analysis was conducted on the dataset size for the test-set used for alignment metric calculation. 49 different fine-tuned-only models were used to calculate representational alignment on the validation sets of Hateful Memes, MM-IMDb and UPMC-Food 101. For each task three models with different seeds were trained, such that the set of models consisted of 15 models for MM-IMDb, 19 on Hateful Memes and 15 on UPMC-Food101. Using the following methodology, we tested whether the dataset size influences the ranks of the alignment metrics:

1. During a forward-pass, hidden activations for both vision and language streams were collected.
2. Alignment metrics (CKA, mkNN, Procrustes, Aligned Cosine) were calculated using these activations.
3. This process was repeated with sample sizes ranging from 64 to 2048 (powers of two).
4. Spearman rank correlation between sample sizes was computed for each model.

Experiments showed high rank correlation across all metrics for sample sizes 64–2048. Table A.1 shows Spearman rank correlations comparing 1024 vs. 2048 samples:

These correlations are substantially higher than for smaller sample sizes (e.g., 512 vs. 2048: CKA $\rho = 0.9646$; Procrustes $\rho = 0.9894$), indicating that 1024 samples provide an excellent compromise between statistical reliability and computational efficiency. These high correlations

Metric	ρ (mean $\pm$ std)	p -value (mean $\pm$ std)
Procrustes	0.9952 ± 0.0150	≤ 0.0000
CKA	0.9771 ± 0.0635	0.0010 ± 0.0080
Aligned Cosine	0.9947 ± 0.0093	≤ 0.0000
mkNN	0.9140 ± 0.1288	0.0071 ± 0.0260

Table A.1: Rank correlation stability for 1024 vs. 2048 samples across metrics.

indicate that metric rankings are stable across sample sizes, validating the choice of 1024 samples for computational efficiency without sacrificing statistical reliability. Notably, Spearman rank correlation (rather than absolute metric values) is consistent across sample sizes, confirming that relative layer-wise comparisons are robust while absolute values are sample-size-dependent, as shown in Figure A.1.

Balancing statistical reliability with computational efficiency, a sample size of 1024 was selected for all further experiments.

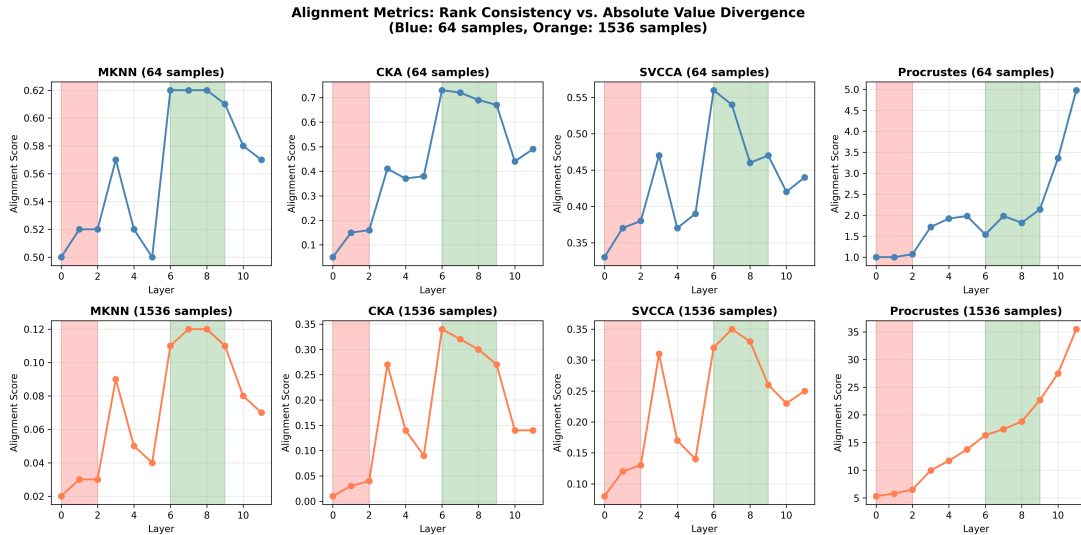


Figure A.1: Rank correlation with absolute values diverging.

A.1.2 Metric Correlation Analysis

Furthermore, preliminarily to this thesis a correlation analysis regarding the number of K in neighborhood measures was conducted with a similar methodology as above. Results showed a high correlation with $\rho > 0.9$ and $p < 10^{-4}$. We therefore chose $k = 32$ for all further experiments.

In addition, correlation studies between metrics were performed. Spearman correlation was employed due to non-normal distributions. In this experiment a wide range of similarity measures was tested. Here layer-wise RAM were collected from models trained only on the three downstream tasks. Results are shown in A.2, indicating redundancies in several metrics: mkNN and Jaccard were functionally identical ($\rho = 1.000$), and CKA with different kernels (linear vs. RBF) correlated at $\rho = 0.999$. In contrast, Procrustes showed weaker correlation with other metrics ($\rho = 0.60 \pm 0.05$), indicating it captures distinct geometric information.

Although mkNN and CKA are highly correlated ($\rho = 0.96, p < 10^{-4}$), these metrics capture different alignment properties. While CKA focuses on global structural correlation, mkNN

emphasizes local neighborhood alignment.

For the final analysis, **Procrustes** and **CKA** were selected as **primary metrics**. In addition, SVCCA and mkNN were chosen for further experiments in order to supplement the primary metrics.

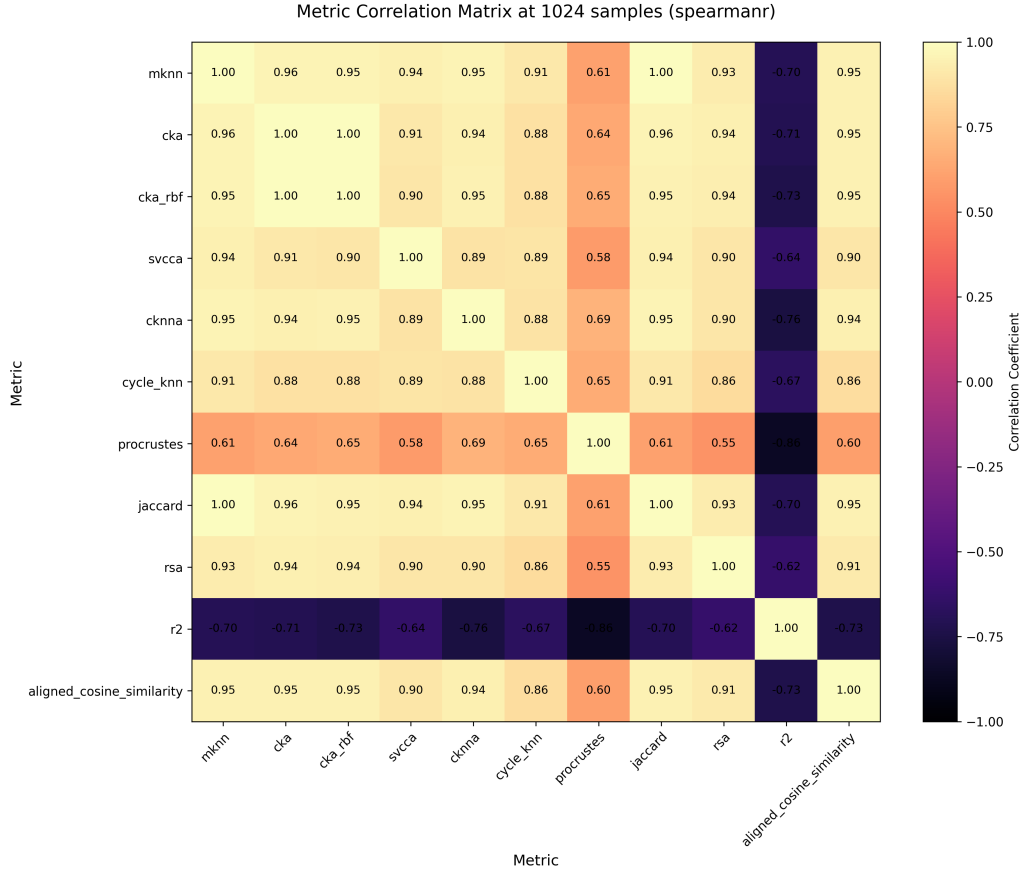


Figure A.2: Correlation Matrix of Metrics: While some metrics show high correlation (e.g., CKA with different kernels, mkNN and Jaccard), Procrustes exhibits low correlation with other metrics, indicating it captures distinct aspects of representational alignment.

While Huh et al. (2024); Tjandrasuwita et al. (2025) performed similar analyses in their paper, the analysis in this thesis was performed in order to show if the results hold for different datasets. Their work focused on more general datasets like Wikipedia-based Image Text Srinivasan et al. (2021), while this thesis focuses on more complex alignment tasks like Hateful Memes. In general, our findings were similar to the ones in their papers, extending their findings for our datasets.

A.2 Architectural Search

To explore asymmetric cross-attention placements, we used Optuna (Akiba et al., 2019) to optimize placement configurations. The search space was parameterized by five variables: the number of cross-attention layers and the mean/spread of placements for each stream. After 72 optimization trials on Hateful Memes and MM-IMDb, we selected the best-performing configurations (O-HM and O-MM in Table 3.2).

However, these optimized architectures did not outperform simple Late Fusion after pretrain-

Parameter	Hateful Memes		MM-IMDb	
	ρ	p-value	ρ	p-value
num_coattn_layers	-0.119	0.348	-0.309	0.013
t_center	0.020	0.874	0.822	<0.001
t_spread	0.218	0.084	-0.021	0.869
v_center	0.138	0.279	0.339	0.006
v_spread	-0.135	0.287	-0.110	0.385

Table A.2: Pearson correlation coefficients (r) and p-values between hyper-parameters and task accuracy.

ing (Table 4.1). Correlation analysis (Table A.2) showed that for MM-IMDb, later text-stream placements correlated with higher accuracy ($\rho = 0.822$, $p < 0.001$), consistent with our main finding that Late Fusion is superior.

Note that this optimization used fine-tuning-only models due to computational constraints. The results suggest that fine-tuning performance does not reliably predict pretrain+finetune performance.

A.3 Alternative Pretraining Approaches

In the pretraining phase, some experiments with only a subset of the three pretraining tasks were conducted. The idea was to see whether multimodal pretraining requires MLM and MIM, or can be performed with Alignment-Prediction (AP) only. It turned out that AP-only failed to converge. Even with sufficiently large training sets (300k of CC, for 4 epochs), no accuracy improvement was observed. The loss remained constant at $0.69 \approx \log(2)$. When adding the two other objectives, training converged successfully as the AP task achieved an accuracy of around 90 %.

In the pretraining phase, also a contrastive loss term for alignment was explored. This was implemented in the AP task, as this was the only pretraining task with cross-modal interaction. The idea was to replace the binary classification task of AP with a contrastive loss between the two modalities. Here the loss was only applied if the two modalities were aligned, as part of this task is that half of the training-set is misaligned. This led to higher accuracies on fine-tuned models in general as can be seen in Figure A.3

However, as contrastive loss directly optimizes for cosine similarity on the corresponding matching pairs, this directly optimizes for representational alignment, with the assumption that higher representational alignment leads to better performance. As this would introduce systematic bias toward higher RAM, we chose not to include it despite its use in recent models. Note that this thesis utilized a contrastive loss in the pretraining phase. As this loss is only aligned on vision without any language interaction, this does not bias the RAM in the same way.

The dual-stream approach for V&L architectures has the drawback that there is no final fused representation. Each modality stream yields a representation coordinated to the other modality (Liang et al., 2023). To utilize all information from multimodal training, the coordinated representations require a fusion step. Three fusion methods were evaluated:

1. Hadamard product: $text \odot image$
2. Summation: $text + image$
3. Concatenation: $[text; image]$

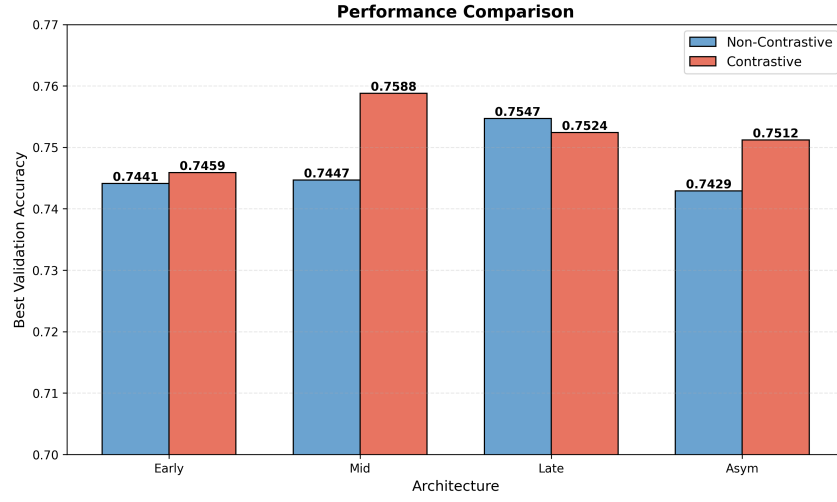


Figure A.3: Hateful Memes: Non-contrastive Pretrain vs. Contrastive Pretrain

Experiments on unpretrained models on Hateful Memes showed similar downstream performance for every method. Summing and Hadamard-product had similar trajectories in terms of train- and validation loss, while concatenation required more epochs to converge.

Appendix B

Detailed Results

This appendix provides supplementary results including statistical tests, layer-wise alignment matrices, and effective dimensionality analysis across all configurations.

B.1 T-Test for Cross-Attention Effect

To statistically test the effect of cross-attention on RAM, we constructed a t-test. The Table B.1 details the results of this experiment. Notable is that the effect is task dependent.

Metric	Mean (No CA)	Mean (CA)	Δ (No CA)	Δ (CA)	p-value
<i>Hateful Memes</i>					
CKA	0.079	0.301	0.014	0.181	<0.001***
mkNN	0.062	0.158	0.003	0.066	<0.001***
SVCCA	0.189	0.419	0.016	0.158	<0.001***
OP	13.38	22.41	1.74	5.12	<0.001***
AUROC	0.646	0.720	0.011	0.031	<0.001***
<i>MM-IMDb</i>					
CKA	0.175	0.634	0.021	0.310	<0.001***
mkNN	0.118	0.373	0.007	0.154	<0.001***
SVCCA	0.266	0.681	0.016	0.188	<0.001***
OP	12.38	15.41	2.80	2.43	0.416
F1-Macro	0.409	0.531	0.056	0.023	0.005**
<i>UPMC-Food101</i>					
CKA	0.126	0.488	0.018	0.196	<0.001***
mkNN	0.115	0.320	0.007	0.104	<0.001***
SVCCA	0.198	0.487	0.023	0.154	<0.001***
OP	14.18	19.31	2.22	3.79	0.002**
Accuracy	0.837	0.897	0.020	0.017	0.582

CA = Cross-Attention; Δ = mean consecutive layer difference; n=117 (No CA), n=27 (CA)

p < 0.01, *p < 0.001

Table B.1: Effect of cross-attention on alignment metrics across all datasets. Cross-attention significantly increases alignment metrics in all tasks. Note: OP shows no significant difference in MM-IMDb; performance difference not significant in UPMC-Food101 at layer-wise granularity.

Table B.2 provides comprehensive performance metrics across all configurations and datasets.

Model	Accuracy	F1-Macro Score	AUROC
<i>Hateful Memes</i>			
Baseline	0.6212 ± 0.0144	0.6076 ± 0.0053	0.6542 ± 0.0016
Early Fusion	0.6807 ± 0.0065	0.6659 ± 0.0043	0.7226 ± 0.0034
Middle Fusion	0.6970 ± 0.0050	0.6720 ± 0.0065	0.7497 ± 0.0090
Late Fusion	0.7019 ± 0.0096	0.6767 ± 0.0159	0.7637 ± 0.0041
Asymmetric Fusion	0.6786 ± 0.0066	0.6490 ± 0.0132	0.7245 ± 0.0018
O-HM	0.6921 ± 0.0048	0.6636 ± 0.0049	0.7460 ± 0.0025
O-MM	0.6848 ± 0.0019	0.6505 ± 0.0049	0.7401 ± 0.0050
Full Cross-Attention	0.6668 ± 0.0025	0.6360 ± 0.0207	0.7074 ± 0.0013
Early-Early Fusion	0.6408 ± 0.0060	0.6219 ± 0.0086	0.6770 ± 0.0093
<i>MM-IMDb</i>			
Baseline	0.9268 ± 0.0005	0.4825 ± 0.0084	0.9011 ± 0.0004
Early Fusion	0.9291 ± 0.0004	0.5118 ± 0.0206	0.9108 ± 0.0002
Middle Fusion	0.9299 ± 0.0002	0.5334 ± 0.0075	0.9145 ± 0.0002
Late Fusion	0.9308 ± 0.0002	0.5448 ± 0.0120	0.9153 ± 0.0005
Asymmetric Fusion	0.9291 ± 0.0001	0.5274 ± 0.0050	0.9103 ± 0.0001
O-HM	0.9296 ± 0.0002	0.5339 ± 0.0081	0.9114 ± 0.0007
O-MM	0.9290 ± 0.0004	0.5340 ± 0.0040	0.9123 ± 0.0003
Full Cross-Attention	0.9278 ± 0.0006	0.5283 ± 0.0036	0.8990 ± 0.0024
Early-Early Fusion	0.9266 ± 0.0005	0.5184 ± 0.0077	0.9038 ± 0.0010
<i>UPMC-Food101</i>			
Baseline	0.8871 ± 0.0012	0.8862 ± 0.0011	/
Early Fusion	0.8930 ± 0.0001	0.8921 ± 0.0001	/
Middle Fusion	0.9181 ± 0.0013	0.9174 ± 0.0014	/
Late Fusion	0.9278 ± 0.0005	0.9272 ± 0.0006	/
Asymmetric Fusion	0.9010 ± 0.0002	0.9001 ± 0.0001	/
O-HM	0.9178 ± 0.0004	0.9170 ± 0.0003	/
O-MM	0.9215 ± 0.0010	0.9209 ± 0.0010	/
Full Cross-Attention	0.9094 ± 0.0010	0.9088 ± 0.0008	/
Early-Early Fusion	0.8998 ± 0.0006	0.8991 ± 0.0007	/

Table B.2: Comprehensive results across all datasets and architectures. For Hateful Memes, AUROC is the primary metric; for MM-IMDb, F1-Macro Score is primary; for UPMC-Food101, Accuracy is primary. All values are mean $\pm$ std across 3 seeds. O-HM represents the best configuration found by Optuna for Hateful Memes, O-MM for MM-IMDb.

B.2 RAM Matrices

These matrices show layer-to-layer CKA similarity. The left plot shows cross-modal alignment (text vs. vision layers), while the middle and right plots show intra-modality consistency. High values in the vision-to-vision plot for Early Fusion (layers 6-12) indicate representational stagnation.

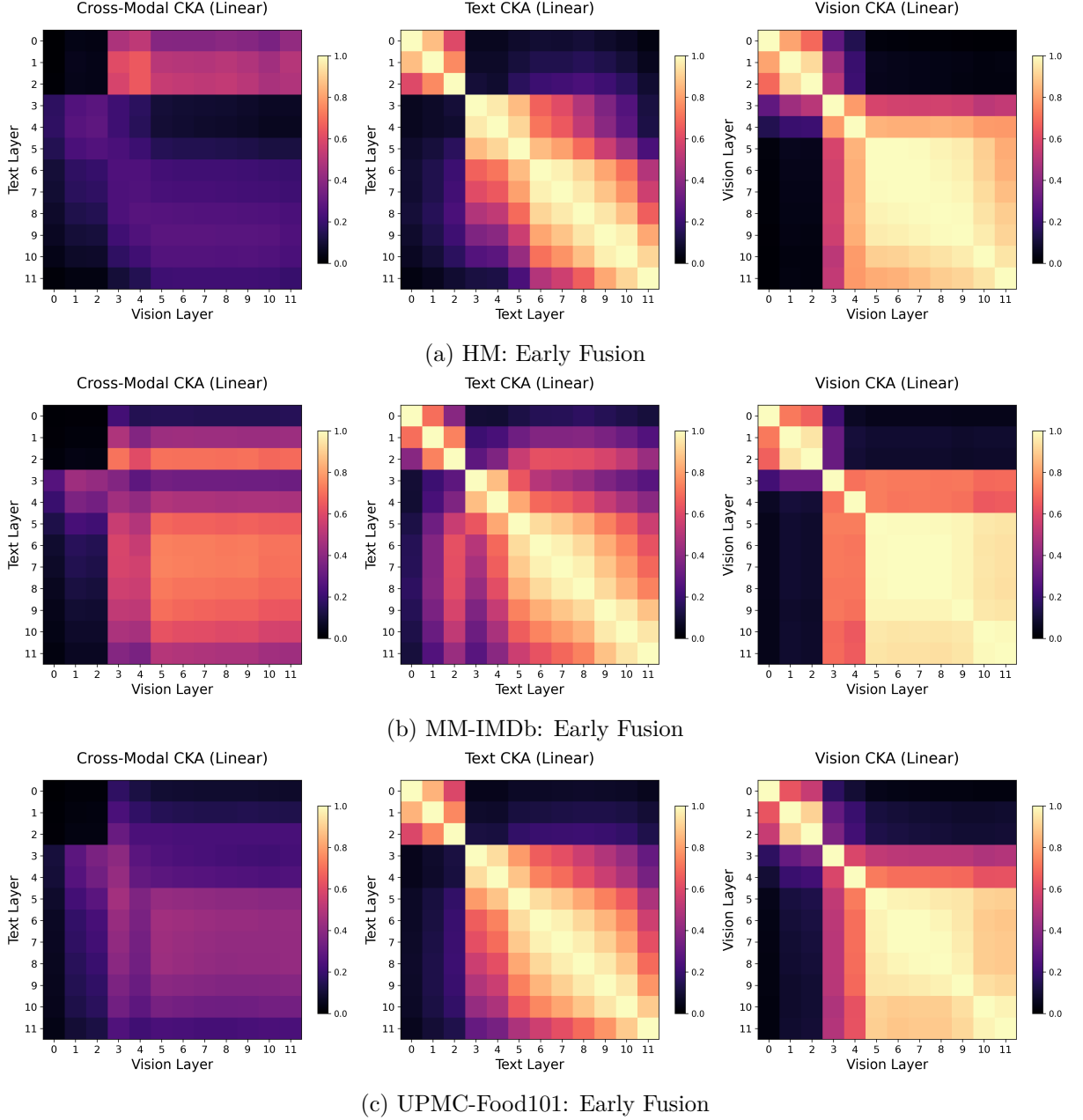


Figure B.1: Detailed representational alignment metrics (CKA) for Early Fusion across the three tasks. Each figure shows three plots. Left: CKA matrix between all layers of both modalities; Middle: Intra-modality similarity for text; Right: Intra-modality similarity for vision.

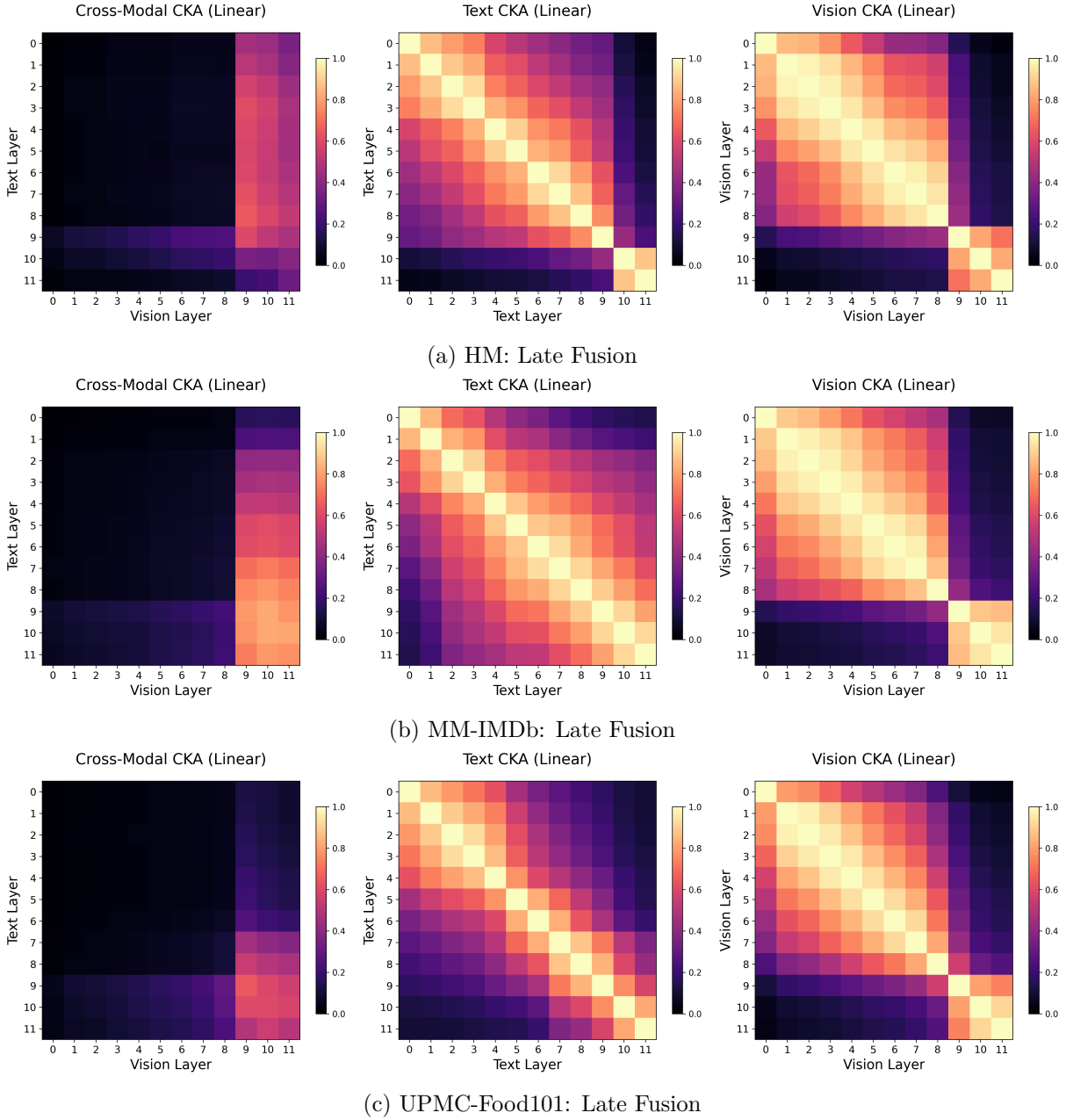


Figure B.2: Detailed representational alignment metrics (CKA) for Late Fusion across the three tasks. Each figure shows three plots. Left: CKA matrix between all layers of both modalities; Middle: Intra-modality similarity for text; Right: Intra-modality similarity for vision.

B.3 Intra-modal CKA Evolution

These plots show how consecutive-layer similarity evolves across the network, revealing where major representational changes occur.

B.4 Effective Dimensionality Results

The computed effective dimensionality of representations has similar trends across all datasets. Up to the cross-modal interaction, the effective dimensionality of the vision stream rises. At

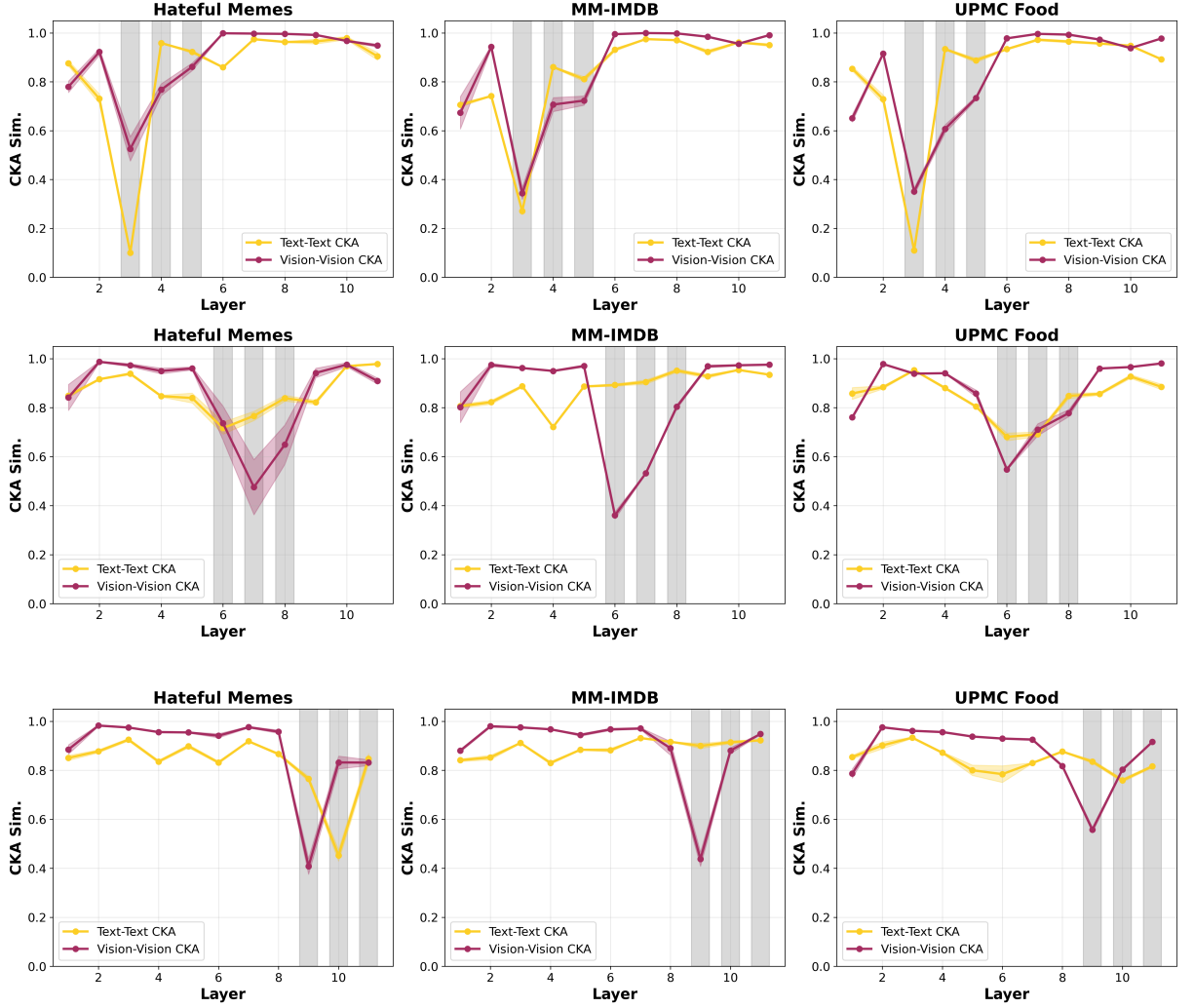


Figure B.3: Intra-modal CKA evolution across layers for Early Fusion (top), Middle Fusion (middle) and Late Fusion (bottom). Each plot shows the CKA trajectory for text (left) and vision (right) separately. The y-axis represents similarity between two consecutive representations $CKA(\mathbf{R}_i, \mathbf{R}_{i+1})$. The x-axis represents the layer index. $n=3$

the stages of cross-attention placements, both streams undergo major restructuring.

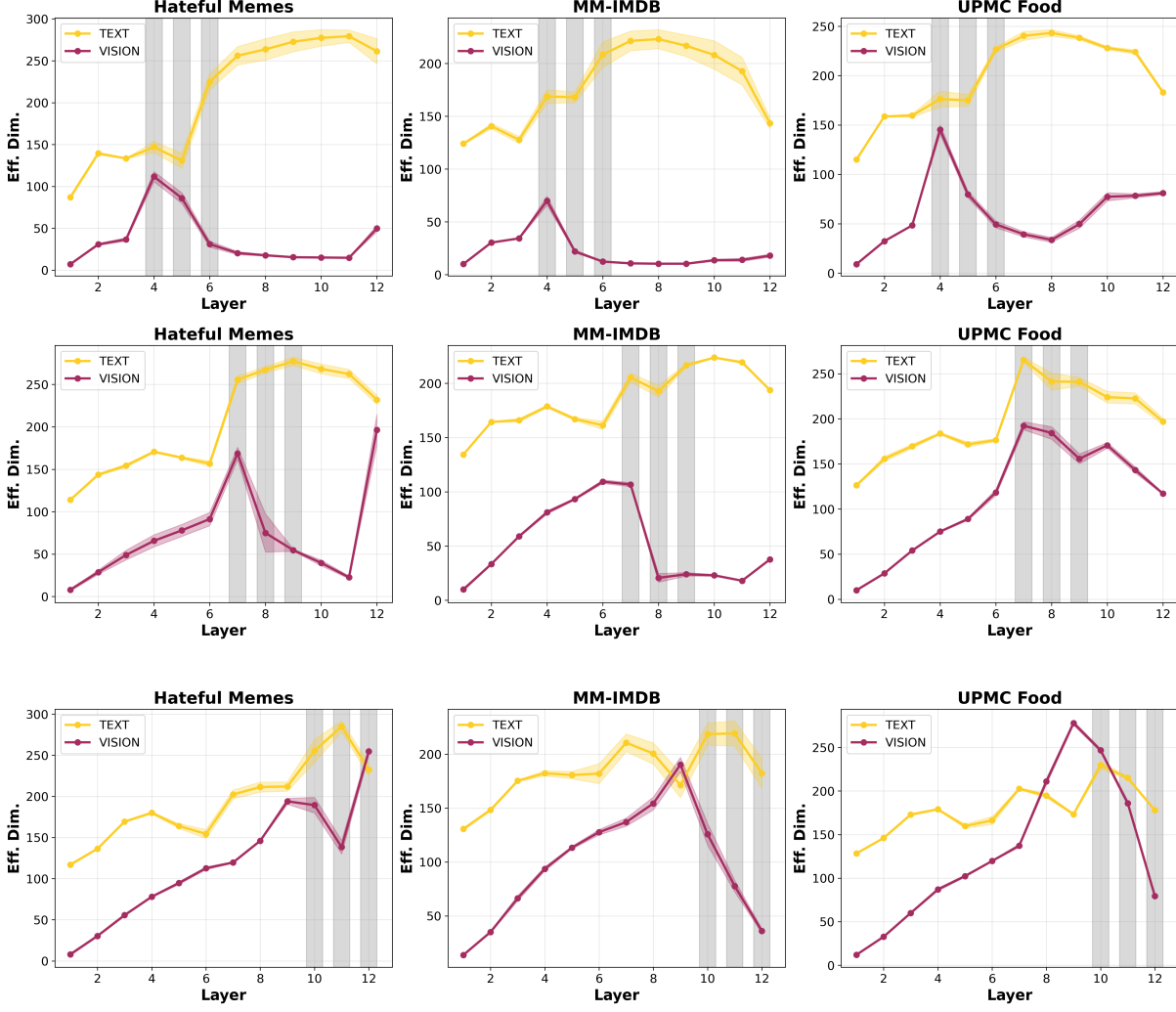


Figure B.4: Effective Dimensionality evolution across layers for Early Fusion (top), Middle Fusion (middle) and Late Fusion (bottom). Each plot shows the effective dimensionality trajectory for text (left) and vision (right) separately. The y-axis represents the effective dimensionality. $n=3$

B.4.1 Baselines

Cross-modal interactions in Early, Middle and Late Fusion induce a dimensionality reduction. However, the baselines show a more nuanced picture. Whilst both configurations show similar trends of effective dimensionality for text, vision differs significantly and helps identify a more general pattern. We refer to Baseline 1 as the model without any cross-attention layers and Baseline 2 as the model with full cross-attention layers.

Baseline 1's trajectory for vision remains limited and never reaches dimensions above 80; for Hateful Memes and MM-IMDb this limitation is even more pronounced (< 50). This directly challenges our hypothesis that cross-attention decreases effective dimensionality, as Baseline 1 shows the lowest effective dimensionality overall. The lack of fusion in the network does not

allow for stable gradient flow between both modalities. Hence, the vision stream never develops any sophisticated features, likely due to the lack of guidance from the text modality.

Baseline 2, on the other hand, is constrained by the necessity to align both modalities at all layers. This leads to coupled representations that differ from the autonomous behavior of Baseline 1. While the effective dimensionality for vision decreases steadily due to this constraint, the final representations remain higher in comparison to Baseline 1, suggesting that the continuous interaction provides structural support.

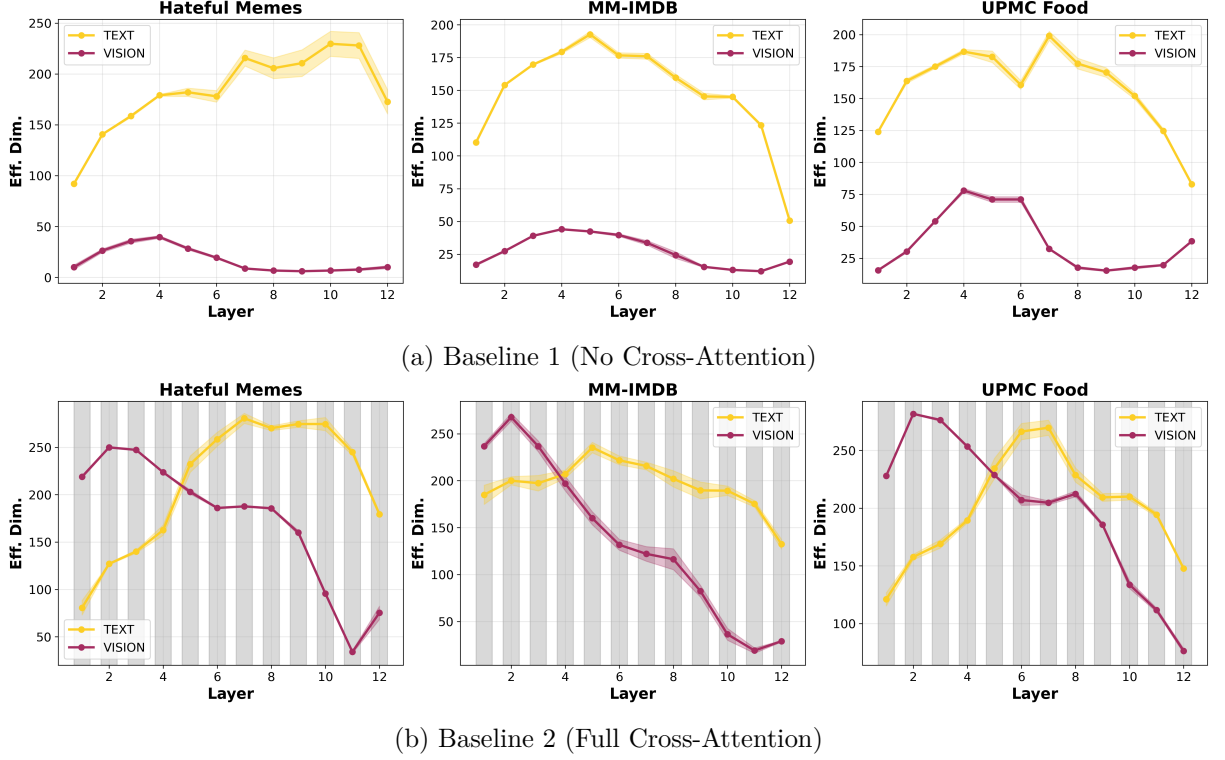


Figure B.5: Effective Dimensionality evolution across layers for Baseline (top) and Full Cross-Attention (bottom). Each plot shows the effective dimensionality trajectory for text (left) and vision (right) separately. The y-axis represents the effective dimensionality, and the x-axis the layer index. $n=3$